Asymmetric Climate Vulnerability and War: Why Empirical Evidence on Extreme Weather Events and Conflict Seems Mixed

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Abstract

Evidence on the effect of climate disasters on armed conflict is mixed. I argue that natural disasters have heterogeneous effects on conflict risk and therefore empirical findings systematically appear mixed. I develop a model of conflict focusing on (i) disaster-induced power shifts and (ii) political groups' relative vulnerability to disasters. It yields two contrasting equilibrium strategies. When two groups have similar vulnerabilities, one opportunistically attacks its rival after a disaster if the latter suffers disproportionately severe damage. Disaster events and conflicts are positively correlated. When one group is inherently more vulnerable, it attacks the relatively robust side preemptively *before* a disaster occurs. Disaster events negatively correlate with conflicts. The theory therefore shows how the same underlying dynamic—disaster-induced power shifts—can generate opposing patterns. I further show that the conditional average treatment effect of disaster events on conflict becomes *negative* in the second case, providing a novel reinterpretation of inconclusive empirical results.

Word count: 10,935

Keywords: Conflict; theoretical implications of empirical models (TIEM); formal political theory; natural disasters; climate change

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1 Introduction

The evidence on the effect of climate anomalies and natural disasters on the risk of violent conflict is mixed. On the one hand, these events can threaten peace by generating resource scarcity, opening a window of opportunity to exploit the affected group that is temporarily vulnerable, or by reducing the opportunity costs of fighting (e.g., Jun and Sethi, 2021; Hsiang et al., 2013; Roche et al., 2020). On the other hand, because sustaining an armed conflict requires belligerents to mobilize material resources, climate shocks can rather reduce the risk of conflict by undermining groups' mobilization and their war-fighting capabilities (Devlin and Hendrix, 2014; Salehyan and Hendrix, 2014). Moreover, some studies report little effect of climate factors on the risk of conflict and find that the positive correlation between them is sensitive to the definition of armed conflict (Buhaug, 2010; Theisen et al., 2011). Mach et al. (2019) also report divergent views among researchers. Hence, it seems plausible to state that there does not exist a wide consensus over the effect of natural disasters and climate events on the risk of armed conflict.[1] Why is the empirical evidence mixed? If natural disasters can lead to armed conflict, *when* do they do so?

In addition to the inconclusive findings and debate, empirical evidence on organized violence seems also "mixed" compared to other types of violence. A meta-analysis by Burke et al. (2024) examines the effect of climate conditions on (i) intergroup violence such as armed conflict, (ii) interpersonal violence, and (iii) self-harm. Among 47 studies on intergroup violence, 34 studies indicate a positive estimated effect of the climate on violence. On the other hand, 23 out of 27 and eight out of nine studies on interpersonal and self-harming violence report positive estimated effects, respectively. Shaver and Bollfrass (2023) also suggests that high temperature affects differently (i) forms of political violence that require collective decision-making and (ii) those that allow individual discretion. Why do the empirical results

[1] For the debates in the empirical literature and overviews of the state of the literature, see, for example, Hsiang et al. (2013); Hsiang and Burke (2014); Buhaug et al. (2014); O'Loughlin et al. (2014); Buhaug (2014); Hsiang and Meng (2014); Mach et al. (2020, 2019).

on *organized* violence, in particular, seem mixed?

I argue that the empirical results in the literature may be divergent because natural disasters affect the risk of conflict in multiple ways, and that such heterogeneous effects *systematically* make the evidence mixed. That is, the theory presented in this paper allows disasters to be positively associated with conflicts in some cases and negatively associated in others. A game-theoretic model shows that two types of contrasting strategy profiles constitute a unique equilibrium in a broad set of parameter values and that these opposing results arise from a single theoretical dynamic: rapid shifts in *military* power between rival political groups due to climate shocks. This dynamic is peculiar to organized armed conflict and does not apply to interpersonal violence and self-harm.

Key concepts that drive the contrasting results are *symmetric* and *asymmetric vulnerabilities* to future natural disasters. It is plausible to assume that the severity of a given disaster is randomly determined and varies across different political groups. For instance, the realized damage from hurricanes depends on each hurricane's course, which is not deterministic. At the same time, however, there is variation in a group's fundamental vulnerability/robustness to disasters, depending on geography, infrastructure, and economic development. For example, some groups in a coastal area may expect to incur disproportionately high costs from tsunamis and hurricanes compared to others located in inland regions. Other groups, such as farmers or pastoralists, may also anticipate that they would be more susceptible to droughts and floods than their rival groups. These groups should be inherently more vulnerable than their political rivals due to the expected asymmetric exposure to an extreme weather event. Hence, different dyads of rival political groups can have different *relative* vulnerabilities.

This observation leads to the following two cases. First, suppose a dyad of rival groups is facing similar levels of disaster risks and hence has symmetric vulnerabilities. Because conflict is costly, the groups in the dyad do not have the incentive to militarily solve a political dispute between them. If a particular climate disaster happens to disproportionately affect one of

the groups, however, the event can harm the military capabilities of the affected group. Conversely, the group that has incurred only a small cost from a disaster may be temporarily advantaged militarily over its political rival. Consequently, the less affected group can commit opportunistic aggression against the other *right after* the extreme weather event. I call this case the *symmetric vulnerability* case.

Second, actual disaster damage on each group in a dyad is *expected* to vary along with their geographical and social resilience. In such a case, the inherently vulnerable side might anticipate that it would become militarily weak after a devastating disaster compared to its relatively robust rival. If it becomes temporarily weak in the presence of a disaster, as in the first case above, its rival might have an opportunistic motive to attack the inherently vulnerable side. Thus, the vulnerable group can have a preemptive motive to attack the rival *before* an actual disaster occurs. Denote this case by the *asymmetric vulnerability* case.

An important implication drawn from the above intuitions is that (i) the actual occurrence of a disaster and armed conflict should be positively correlated in the first case (because war follows the asymmetric post-disaster balance of power) but (ii) they should be negatively associated in the second (because war erupts in the absence of a disaster). Moreover, based on the game-theoretic model, I show that the causal effect of disasters on conflict in the sense of the conditional average treatment effect (CATE) becomes *negative* in the second case. This observation provides a novel theoretical interpretation of why the empirical evidence on climate conflict looks mixed: I argue that overlooking the second mechanism (i.e., war caused by expected asymmetric exposure to disasters) may lead researchers to understate the true effect of climate anomalies by possibly misreading the preemptive motive as a stabilizing effect of disasters. Thus, the theory suggests that climate shocks are conflict-inducing in a broader sense because disaster *risk* also drives preemptive conflict. I also revisit the meta-analysis by Burke et al. (2024) to conduct a suggestive empirical exercise and illustrate an implication drawn from the asymmetric-vulnerability theory.

1.1 Contributions to the Literature

This study contributes to the literature on extreme weather events and armed conflict in mainly three ways.[2] First, perhaps most importantly, this paper is an example of the growing Theoretical Implications of Empirical Models (TIEM) approach applied to the climate-conflict nexus. Studies in TIEM aim to theoretically comprehend empirical findings (e.g., Ashworth et al., 2021; Slough, 2023; Bueno de Mesquita and Tyson, 2020; Wolton, 2019, 2021; Slough, 2024; Fu and Slough, 2025; Marshall, 2024). I take this approach by formally pointing out how climate disasters can pose heterogeneous treatment effects on the risk of conflict, and empirical results can look mixed. While some recent studies, such as Ide (2023) and Nemeth and Lai (2022), find that disasters can induce both escalation and de-escalation, the present paper provides a novel reinterpretation of the mixed empirical evidence on the causal effects of natural disasters and extreme weather events on violent conflict. Not only does the paper present a simple model consistent with divergent empirical findings, but also a simple empirical exercise based on studies in the meta-analysis by Burke et al. (2024), though not conclusive, helps us reconcile the debate theoretically.

Second, this paper proposes a *military* theory of natural disasters and conflict. In contrast to the present paper and notable exceptions such as Bas and McLean (2021), Eastin (2016), Salehyan and Hendrix (2014), and Zhao (2026), there are relatively few studies that emphasize the military impacts of natural disasters, whereas theories examining the effects of climate shocks on conflict tend to focus on their *economic* consequences. The comprehensive review by (Burke et al., 2024, 252) classifies the major possible mechanisms through which climate conditions influence conflict into five categories, none of which is directly related to their military impacts: "(1) economic conditions, income, and agricultural productivity; (2) socio-demographic factors; (3) migration and transportation costs; (4) policy, politics and institutions; and (5) psychological and physiological factors." While armed conflicts are

[2]For review, see Buhaug and von Uexkull (2021); Burke et al. (2015); Koubi (2019); Burke et al. (2024).

military events, the number of studies focusing on the military impacts of climate events on military phenomena is relatively limited.

As a prominent example of studies examining the economic mechanism, Chassang and Padró i Miquel (2009) have been a canonical model adopted by scholars of climate anomalies and conflict (e.g., Baysan et al., 2019; Roche et al., 2020; Burke et al., 2015). Their model of conflict under economic shocks assumes that (i) an external shock affects players' resources symmetrically and that (ii) a player who unilaterally attacks the other has an offensive advantage, where the attacker prevails with probability greater than one-half.[3] When the current economic production is small, players can have an incentive to attack the other because of a temporarily lower opportunity cost of fighting.

It is important to point out that the presence of an offensive advantage is a *military* assumption. The assumption that there is a first-mover advantage in war is not directly related to the impact of the economic shock. Moreover, as the models of Chassang and Padró i Miquel (2009) and the subsequent studies implicitly assume that both players incur the same damage, the probability of victory in war is constant regardless of the effect of the shock. In fact, an offensive advantage is a *necessary condition* for war in the canonical model of Chassang and Padró i Miquel (2009) and its application to climate change by Roche et al. (2020).[4] In sum, the prominent *economic* explanation drawing on the opportunity cost logic rests on a *military* assumption.

It follows that scholars need to incorporate the military aspect of climate shocks, which is peculiar to organized violence as opposed to interpersonal violence and self-harm. To this end, I conceptualize natural disasters as negative random shocks to political groups' military capabilities. On the one hand, responding to a disaster requires the government to reallocate

[3]For studies that examine differential (rather than uniform) effects of economic shocks on conflict, see Dube and Vargas (2013) and McGuirk and Burke (2020).

[4]Conflict can occur in equilibrium in Baysan et al. (2019) without the offensive advantage assumption when a psychological gain from violence is large enough. However, this psychological payoff is an individual component, which is more relevant to interpersonal violence rather than organized violence. See also Shaver and Bollfrass (2023) for psychological effects of climate conditions as a possible source of the difference between interpersonal and organized violence.

resources, thereby reducing its counterinsurgency capabilities (Eastin, 2016). Moreover, if a disaster destroys road networks, the government might have lower state capacity and deteriorated access to peripheral regions. When this is the case, rebel groups in those regions can have a higher chance of successful insurgency (e.g., Müller-Crepon et al., 2021; Fearon and Laitin, 2003). On the other hand, rebel groups can also face a severe scarcity of resources that renders recruitment more challenging (Walch, 2018). To incorporate these military impacts of disasters, the model presented below directly connects disaster damage and military capability in a given period.

Third, in terms of the concept of relative vulnerabilities to future disasters, one of the closest studies is Bas and McLean (2021). They find that conflicts are associated with disaster *risks* and, thus, that the expectations of natural disasters can induce preemptive attacks, which is consistent with the asymmetric vulnerability case discussed below. The present model builds on their insight and further allows for recurrent disasters, the recovery from disaster damage, and asymmetry in realized damage. These extensions unify two channels within a single framework: opportunistic conflict triggered by realized post-disaster shifts in the balance of power and preemptive conflict driven by expected asymmetric vulnerability to future disasters. In this sense, the paper also contributes to the broader literature on conflict under rapid shifts in power (Fearon, 2004; Powell, 2004, 2006) by showing how sudden power fluctuations can generate both positive and negative correlations between external shocks and armed conflict.[5]

[5]See also Benson and Smith (2023), Krainin et al. (2022), Little and Paine (2023), Sawada (2024), Smith (2019), and Wolford and Masumura (2025) for recent models of shifts in power and conflict. In particular, Little and Paine (2023) is closely related to the present model because they also assume that the probability that one group wins in war is randomly drawn each period. Smith (2019) shows how arms transfers from an ally can mitigate power fluctuations.

2 The Model

This paper develops a simple infinite horizon game that incorporates the relative vulnerability to future disasters. A key feature is that asymmetric exposure is a necessary condition for conflict in the model. Thus, war on any equilibrium path is triggered by the actual occurrence or expectation of natural disasters that can impose disproportionate damage on one of the players.

2.1 Setup

Two players (e.g., rival political groups in the context of domestic conflict or neighboring states in the context of interstate conflict) 1 and 2 interact each period $t = 1, 2, ...$ In each period, players have baseline resources whose amount, θ, is constant across players and periods. At the beginning of each period, Nature determines the presence/absence of a natural disaster as a draw of an identically and independently distributed random variable D_t following a Bernoulli distribution with mean π: in each t, a disaster occurs (i.e., $D_t = 1$) with probability π and does not (i.e., $D_t = 0$) with probability $1 - \pi$.[6] We model asymmetric exposure to a disaster as another independently and identically distributed random variable Z_t over the unit interval with mean μ. Suppose that (i) Z_t follows a distribution with a continuously differentiable cumulative distribution function F whose density is f and that (ii) it has a full support over $[0, 1]$. In a disaster period t, the materialized value of Z_t, denoted as z_t, discounts (i) the resources of Player 1 to $z_t\theta$ and (ii) those of Player 2 to $(1 - z_t)\,\theta$. Namely, after $D_t = 1$, Nature determines the level of asymmetry in the damage imposed on each player.[7]

After Nature's moves, players observe the state (D_t, Z_t) and simultaneously decide whether

[6] We call the period in which a natural disaster has occurred a "disaster period." Similarly, we label a period without a disaster as a "no-disaster period".

[7] Assume that Nature does not draw Z_t when $D_t = 0$. An alternative assumption is that Nature draws $Z_t \in [0, 1]$ regardless of D_t. Because the relative disaster damage does not appear when there is no disaster, the difference in the assumption does not affect the results.

to attack the other or not. In this model, we assume that a conflict is a unilateral act: war ensues when at least one player chooses to fight. If neither player fights, players receive a flow payoff equal to the amount of their wealth in the period (θ when there is no disaster or $z_t\theta$ and $(1-z_t)\,\theta$ when a disaster has erupted in the period), and the game proceeds to the next period.

When either player attacks the opponent, a game-ending war ensues. Its outcome is determined by the amount of players' resources: the probability that player i wins a war is

$$\begin{aligned}
p_i^{\neg \mathtt{d}} &\equiv \frac{\theta}{2\theta} \\
&= \frac{1}{2} \quad \text{(when a disaster did not occur); or} \\
p_1^{\mathtt{d}}(Z_t) &\equiv \frac{Z_t\theta}{Z_t\theta + (1-Z_t)\,\theta} \\
&= Z_t \quad \text{(when a disaster occurred in } t\text{) and} \\
p_2^{\mathtt{d}}(Z_t) &\equiv 1 - p_1^{\mathtt{d}}(Z_t) \\
&= 1 - Z_t.
\end{aligned}$$

Because a war is costly, it discounts the amount of resources in the society by $c \in (0,1)$: the remaining resource is $\theta(1-c)$ or $2\theta(1-c)$ when war erupts in disaster and no-disaster periods, respectively. If a war ensues, the game ends and the winner receives all remaining resources in the society in the current period ($\theta(1-c)$ or $2\theta(1-c)$) and a long-term benefit from its dominance over resources. By a long-term benefit, we mean that the victor gains the entire pie 2θ that can be discounted in half with probability π for infinitely many periods.

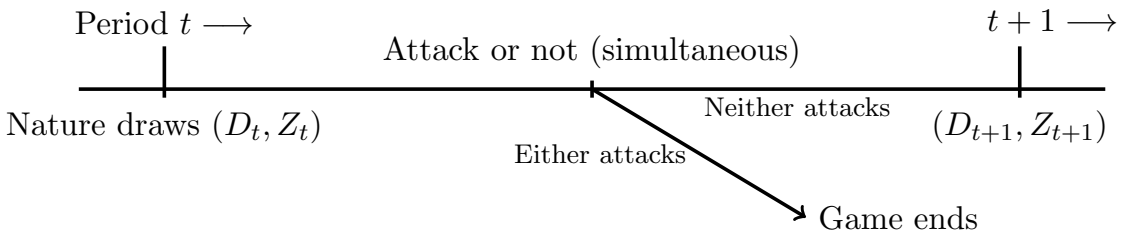


Figure 1: Timeline

Formally, the long-term payoff from dominance is:

$$\begin{aligned}\overline{\mathcal{V}} &\equiv \mathbb{E}_{D_t,Z_t}\left[\sum_{t=1}^{\infty}\delta^{t-1}\left(D_t\left(Z_t\theta+(1-Z_t)\,\theta\right)+(1-D_t)\,2\theta\right)\right] \\ &= \sum_{t=1}^{\infty}\delta^{t-1}\left[\pi\theta+(1-\pi)\,2\theta\right] \\ &= \frac{\theta(2-\pi)}{1-\delta},\end{aligned}$$

where $\delta \in (0,1)$ is a common discount factor, and we assume that it is sufficiently large: $\frac{\delta}{1-\delta}(2-\pi) > c$.

Figure 1 summarizes the timeline of the game. We assume complete information. To focus on interesting cases in which disasters affect players' equilibrium strategies, we introduce the following assumption that excludes trivial cases, such as an equilibrium in which players attack the opponent after any history.

Assumption 1. Players do not attack the other one when they are indifferent between attacking and not attacking.

This assumption also enables us to focus on pure strategies.

Our solution concept is pure-strategy (stationary) Markov perfect equilibrium (MPE).[8] That is, a strategy in this model is a mapping from the state space $\{0,1\}\times[0,1]$ to the action space $\{\texttt{Attack},\texttt{Not}\}$.

[8]Whereas we focus on MPE in the analysis section, we use subgame perfect equilibrium for the preliminary results (Remarks 1 and 2) presented in the next section.

2.2 Comments on the Model

Before analyzing the model, it is useful to discuss the interpretations of its key features. First, the model presents a *military* theory of natural disasters and conflict. Namely, the random variable Z_t represents the conceptualization of a disaster as a random negative shock to players' military capabilities. Hence, unlike the climate-conflict applications of Chassang and Padró i Miquel (2009), such as Baysan et al. (2019) and Roche et al. (2020), a climate shock directly translates into a shift in the balance of power in two steps. Firstly, the realized damage of a disaster z_t discounts the players' resources. Secondly, the affected amounts of resources determine the outcome of conflict via the probability of victory $p_i^{\mathtt{d}}(z_t)$, which is either z_t or $1 - z_t$.

Second, the key parameter $\mu \equiv \mathbb{E}[Z_t]$ represents the *relative vulnerabilities* of the players to future disasters. One interpretation is that it captures the region-specific vulnerabilities that amplify the costs caused by natural disasters. For example, coastal areas may be more vulnerable to hurricanes (due to storm surges) and post-earthquake tsunamis. Some mountainous regions may also face a higher risk of landslides after excessive rainfall or earthquakes. In the model, μ close to $1/2$ implies that the two groups have similar vulnerabilities to natural disasters (the symmetric vulnerability case). On the other hand, a small or large value of μ means that one of the players is more vulnerable to climate risks (the asymmetric vulnerability case). A specific draw of Z_t, i.e., denoted as z_t, represents the *realized* exposure to a disaster, which varies depending on which region the disaster hits most severely.

Third, the realized disaster damage z_t also determines players' flow payoffs after a disaster period. There are at least two interpretations. For one, as discussed above, military capabilities may reflect economic resources: a disaster can negatively affect the economic resources $z_t\theta$ and $(1 - z_t)\theta$, which, in turn, translate into military power. Conversely, economic gains may reflect the balance of military capabilities: one can also consider the flow payoffs as a product of bargaining between rival groups, where the stronger side obtains more

resources. Thus, while the model in the main text does not have a bargaining stage, one can think of the flow payoffs as a result of a bargaining process that allocates economic resources commensurate with each player's military capability.[9]‘

Fourth, the complete-information model implies that the distribution F and, in particular, its mean μ are common knowledge, which establishes a scope condition of this model. Namely, we focus on relatively foreseeable disasters because the complete-informations assumption means that players can evaluate and take into account future disaster risks. Examples include seasonal climate anomalies such as storms, floods, and droughts. On the other hand, other natural disasters that are difficult to anticipate, including earthquakes, are beyond the scope.

Fifth, the negative impact of disasters on social welfare is fixed. Namely, any extreme weather event always halves the resources in the society composed of players 1 and 2. While the distribution of discounted resources varies according to Z_t, each disaster cannot make the total amount of resources more or less than 1/2. This simplifying assumption makes clear how *asymmetric* exposure to extreme weather events, rather than their absolute magnitude itself, affects players' incentive for maintaining intergroup peace and gambling on a costly war.

3 Analysis

This section presents two different equilibrium strategies in which realized and expected asymmetric costs of natural disasters trigger armed conflict. In the first, symmetric vulnerability case, a player can attack the other to exploit the opponent's temporary vulnerability arising from the latter's disproportionate damage from a disaster in the current period. In the second, asymmetric vulnerability case, a player has the incentive to fight in a no-disaster

[9] Appendix C shows that the key insights of the model remain even when we allow an arbitrary bargaining protocol between Nature's move and players' decision about whether to fight.

period when she expects she will become disproportionately vulnerable in the future.[10]

3.1 Benchmark without Asymmetric Exposure to Disasters

Before presenting the two cases, I introduce the following preliminary results. They show that the possibility of a disaster with asymmetric negative impacts is a necessary condition for war in this model. That is, the game always proceeds peacefully without (i) the risk of natural disasters and (ii) asymmetric exposure to them. These remarks highlight the difference between the present model and the canonical model by Chassang and Padró i Miquel (2009), where both players experience the same economic shock.

Remark 1. *Suppose $\pi = 0$. Then, under Assumption 1, the unique subgame perfect equilibrium (SPE) is a strategy profile where both players never attack the opponent.*

Remark 2. *Suppose $\pi \in (0, 1)$ but $z_t = 1/2$ for any t. That is, the damage from a disaster is always symmetric. Then, under Assumption 1, the unique SPE is a strategy profile where both players never attack the opponent.*

These results are straightforward. Consider Remark 1. Because war is costly and there is no asymmetric information or a sudden power shift, a unilateral attack cannot be optimal after any history, which is in line with standard models of armed conflict (Fearon, 1995). The logic behind Remark 2 is also intuitive: the complete symmetry in exposure to a disaster prevents the temporary advantage that incentivizes aggression.[11] The model in which war can occur only under some risk of natural disasters with asymmetric exposure allows us to theoretically identify the effects of those extreme events on the risk of armed conflict.

[10] Appendix B shows (i) the existence of an equilibrium of the game and (ii) the condition under which there exists a unique optimal cutoff strategy in disaster periods in each strategy profile that can constitute an equilibrium. Below, to have unique cutoffs, assume that the density f satisfies the condition specified in Proposition 6 in the appendix.

[11] In the model of Chassang and Padró i Miquel (2009), war erupts under symmetric economic shocks because they assume the presence of offensive advantage: if only one player launches a unilateral attack, her probability of victory is greater than 1/2, which renders the outcome more conflict-prone. The war outcome in the present model is determined by the asymmetry in resources rather than by who started the war.

3.2 Symmetric Vulnerability and the Opportunistic Motive for War

I present an equilibrium in which a conflict erupts as a result of opportunistic aggression by a player who is temporarily advantaged in a disaster period. To derive the equilibrium, we consider the following stationary strategies:

(S-I): In a disaster period, i.e., t such that $D_t = 1$, either one of the players attacks the other if the realized cost of the disaster is disproportionately incurred.

- When z_t is larger than a threshold $\overline{z}^{\mathtt{S}}(\mu)$ given μ, Player 1 attacks 2.
- When z_t is smaller than a threshold $\underline{z}^{\mathtt{S}}(\mu)$ given μ, Player 2 attacks 1.

(S-II): In a no-disaster period, i.e., t such that $D_t = 0$, neither player fights.

Slightly abusing notation, we often write the threshold values as $\overline{z}^{\mathtt{S}}$ and $\underline{z}^{\mathtt{S}}$ instead of $\overline{z}^{\mathtt{S}}(\mu)$ and $\underline{z}^{\mathtt{S}}(\mu)$, which are functions of μ as we discuss below.

When we assume the players follow the above strategies, on the equilibrium path, a war occurs if the *realized* disaster damage in the current period is highly asymmetric. The following result specifies the condition under which they constitute an equilibrium.

Proposition 1 (Symmetric vulnerability)**.** *Consider an interval $M^{\mathtt{S}} \equiv \left[\underline{\mu}^{\mathtt{S}}, \overline{\mu}^{\mathtt{S}}\right]$, where the thresholds $\underline{\mu}^{\mathtt{S}}$ and $\overline{\mu}^{\mathtt{S}}$ are defined below. When $\mu \in M^{\mathtt{S}}$, the strategies* (S-I) *and* (S-II) *constitute an MPE.*

- *Given $\mu \in M^{\mathtt{S}}$, the thresholds above/below which one player attacks the other when*

$D_t = 1$ *are given as the solution to the following system of equations.*

$$\begin{cases} \overline{z}^{\mathtt{S}}(\mu) = & \delta \cdot \dfrac{(1-\pi) + \pi\left(\mu + \left(\displaystyle\int_0^{\underline{z}^{\mathcal{S}}(\mu)} z\mathrm{d}F(z) + \int_{\overline{z}^{\mathcal{S}}(\mu)}^1 z\mathrm{d}F(z)\right)\left[\delta\dfrac{2-\pi}{1-\delta} - c\right]\right)}{\left(1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathcal{S}}(\mu), \underline{z}^{\mathcal{S}}(\mu)\right)\right)\right)\right)\left[\delta\dfrac{2-\pi}{1-\delta} - c\right]} \\ \underline{z}^{\mathtt{S}}(\mu) = & 1 - \delta \cdot \dfrac{(1-\pi) + \pi\left(1 - \mu + \left[\begin{array}{l} 1 - \Delta_F\left(\overline{z}^{\mathcal{S}}(\mu), \underline{z}^{\mathcal{S}}(\mu)\right) \\ -\left(\displaystyle\int_0^{\underline{z}^{\mathcal{S}}(\mu)} z\mathrm{d}F(z) + \int_{\overline{z}^{\mathcal{S}}(\mu)}^1 z\mathrm{d}F(z)\right)\end{array}\right]\left[\delta\dfrac{2-\pi}{1-\delta} - c\right]\right)}{\left(1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathcal{S}}(\mu), \underline{z}^{\mathcal{S}}(\mu)\right)\right)\right)\right)\left[\delta\dfrac{2-\pi}{1-\delta} - c\right]}, \end{cases} \tag{1}$$

where $\Delta_F(b,a) \equiv F(b) - F(a)$ *with* $a \leq b$.

- *The maximum and minimum of* $M^{\mathcal{S}}$ *are, respectively, given as part of the solution to a system of three equations composed of (i) each of the following equations and (ii) two equations in (1) with* $\mu = \overline{\mu}^{\mathcal{S}}$ *or* $\mu = \underline{\mu}^{\mathcal{S}}$. *Letting* F_μ *denote the CDF of* Z_t *with mean* μ, *the thresholds in* μ *are implicitly specified by*

$$\begin{aligned} \overline{\mu}^{\mathcal{S}} &= \frac{1}{2} + \delta\frac{2-\pi}{1-\delta}\left(\int_0^{\underline{z}^{\mathcal{S}}\left(\overline{\mu}^{\mathcal{S}}\right)} \frac{1}{2} - z\mathrm{d}F_{\overline{\mu}^{\mathcal{S}}}(z) + \int_{\overline{z}^{\mathcal{S}}\left(\overline{\mu}^{\mathcal{S}}\right)}^1 \frac{1}{2} - z\mathrm{d}F_{\overline{\mu}^{\mathcal{S}}}(z)\right) \\ &\quad + c\left(\int_0^{\underline{z}^{\mathcal{S}}\left(\overline{\mu}^{\mathcal{S}}\right)} z\mathrm{d}F_{\overline{\mu}^{\mathcal{S}}}(z) + \int_{\overline{z}^{\mathcal{S}}\left(\overline{\mu}^{\mathcal{S}}\right)}^1 z\mathrm{d}F_{\overline{\mu}^{\mathcal{S}}}(z) + \frac{1-\delta}{\delta\pi}\right) \quad \textit{and} \\ \underline{\mu}^{\mathcal{S}} &= \frac{1}{2} + \delta\frac{2-\pi}{1-\delta}\left(\int_0^{\underline{z}^{\mathcal{S}}\left(\underline{\mu}^{\mathcal{S}}\right)} \frac{1}{2} - z\mathrm{d}F_{\underline{\mu}^{\mathcal{S}}}(z) + \int_{\overline{z}^{\mathcal{S}}\left(\underline{\mu}^{\mathcal{S}}\right)}^1 \frac{1}{2} - z\mathrm{d}F_{\underline{\mu}^{\mathcal{S}}}(z)\right) \\ &\quad - c\left(\int_0^{\underline{z}^{\mathcal{S}}\left(\underline{\mu}^{\mathcal{S}}\right)} 1 - z\mathrm{d}F_{\underline{\mu}^{\mathcal{S}}}(z) + \int_{\overline{z}^{\mathcal{S}}\left(\underline{\mu}^{\mathcal{S}}\right)}^1 1 - z\mathrm{d}F_{\underline{\mu}^{\mathcal{S}}}(z) + \frac{1-\delta}{\delta\pi}\right). \end{aligned}$$

We denote this equilibrium by $(\mathtt{S})$ as in symmetric vulnerability. Note that war can erupt only in disaster periods on the equilibrium path. That is, peace sustains as long as extreme asymmetry in players' resources does not arise from a natural disaster. Conversely, war occurs when highly disproportionate exposure to an extreme weather event generates an opportunistic motive of the temporarily advantaged player. Thus, the equilibrium probability of conflict is (i) zero in a given no-disaster period and (ii) positive in a disaster period.

This result is intuitively straightforward. When μ is in the interval M^{S}, unlike the second case presented below, the players' inherent vulnerabilities to natural disasters are relatively symmetric. This means that military parity between the players is likely to remain even in the event of a disaster. Because, on average, military parity is unlikely to fluctuate under an extreme weather event, neither player obtains a temporary military advantage (and an opportunistic motive for war) unless such a balance is disturbed by a disaster that imposes disproportionately high costs on one of them. We shall come back to this point to compare this equilibrium and the one introduced below.

The key assumption sustaining the above equilibrium is that the economy of the society will recover from the negative economic shock caused by destructive disasters in the next period (Chassang and Padró i Miquel, 2009). If post-disaster humanitarian relief helps the players rebuild their economy and military capabilities, the one who incurred disproportionately *small* damage faces a closing window of opportunity: she has the incentive to attack the temporarily vulnerable rival before parity in players' military power is restored.

Figure 2 provides an example. The horizontal axis and vertical axis represent the value of $\mathbb{E}[Z_t] = \mu$ and the density of Z_t, respectively. When $c = 0.55$, $\delta = 0.5$, $\pi = 0.1$, and $Z_t \sim \text{Beta}(10, 10)$ (i.e., $\mu = 0.5$, the solid vertical line in the figure), we have $\underline{z}^{\mathsf{S}}(\mu = 0.5) \approx 0.3$ (dotted line) and $\overline{z}^{\mathsf{S}}(\mu = 0.5) \approx 0.7$ (dashed line). Intuitively, on this equilibrium path, each player attacks the other in period t such that $D_t = 1$ if she has at least a seventy percent chance of winning. To compare the results below, note again that neither has the incentive to fight in a no-disaster period.

Indeed, one can find examples of this dynamic in existing studies. Jun and Sethi (2021) focus on historical episodes of invasion in ancient Korea. They find that kingdoms affected by climate shocks (drought, heavy rainfall, heavy snow, or frost) were more likely to be the target of invasion in the year. This is consistent with the symmetric vulnerability case, where one of the geographically proximate kingdoms became vulnerable and the target of an

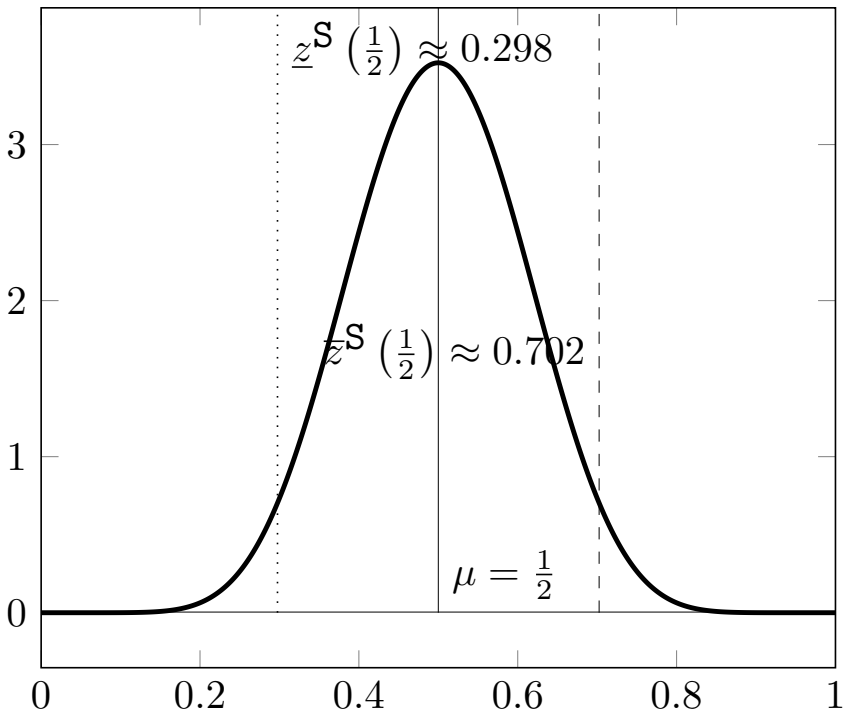


Figure 2: Thresholds in (S) ($c = 0.55, \delta = 0.5, \pi = 0.1, Z_t \sim \text{Beta}(10, 10)$)

opportunistic attack from a neighboring rival kingdom. Kikuta (2019) considers post-disaster reconstruction as recovery from disaster damage that shifts the balance of power back to the pre-disaster level. Leveraging data on the 2004 tsunami in Sri Lanka, Kikuta (2019) shows that the number of post-tsunami housing reconstructions led to conflict.

3.3 Asymmetric Vulnerability and the Preemptive Motive for War

Next, we consider the conditions under which a war occurs due to an inherently vulnerable player's fear of future exploitation. To present equilibria that illustrate this case, consider the following stationary strategies.

(A-I): In a disaster period, i.e., t such that $D_t = 1$, either one of the players attacks the other if the realized cost of the disaster is disproportionately incurred.

- When z_t is larger than a threshold $\overline{z}^{\mathtt{A}}(\mu)$, Player 1 attacks 2.

– When z_t is smaller than a threshold $\underline{z}^{\mathtt{A}}(\mu)$, Player 2 attacks 1.

(A-II): In a no-disaster period, i.e., t such that $D_t = 0$, the more vulnerable player attacks the other.

In the strategies above, notice that war immediately ensues if there is no disaster. The following proposition provides the conditions under which (A-I) and (A-II) constitute two equilibria with a different player who attacks at a no-disaster period.

Proposition 2 (Asymmetric vulnerability)**.** *Consider the union of two half-open intervals* $M^{A} \equiv \left[0, \underline{\mu}^{A}\right) \cup \left(\overline{\mu}^{A}, 1\right]$*, where the thresholds* $\underline{\mu}^{A}$ *and* $\overline{\mu}^{A}$ *are defined below. When* $\mu \in M^{A}$*, the strategies* (A-I) *and* (A-II) *constitute an MPE.*

- *In no-disaster periods, Player 1 attacks Player 2 when* $\mu < \underline{\mu}^{A}$ *and Player 2 attacks Player 1 when* $\mu > \overline{\mu}^{A}$*.*
- *Given* $\mu \in M^{A}$*, the thresholds* $\overline{z}^{A}(\mu)$ *and* $\underline{z}^{A}(\mu)$ *are given as the solution to the following system of equations.*

$$\begin{cases} \overline{z}^{A}(\mu) = & \delta \cdot \dfrac{\left[\begin{array}{l}(1-\pi)\left[\dfrac{1-\delta\pi/2}{1-\delta} - c\right] + \\ \pi\left(\mu + \left(\displaystyle\int_0^{\underline{z}^{A}(\mu)} z\mathrm{d}F(z) + \int_{\overline{z}^{A}(\mu)}^{1} z\mathrm{d}F(z)\right)\left[\delta\dfrac{2-\pi}{1-\delta} - c\right]\right)\end{array}\right]}{\left(1 - \delta\pi\Delta_F\left(\overline{z}^{A}(\mu), \underline{z}^{A}(\mu)\right)\right)\left[\delta\dfrac{2-\pi}{1-\delta} - c\right]} \\ \underline{z}^{A}(\mu) = & 1 - \delta \cdot \dfrac{\left[\begin{array}{l}(1-\pi)\left[\dfrac{1-\delta\pi/2}{1-\delta} - c\right] + \\ \pi\left(1 - \mu + \left[\begin{array}{l}1 - \Delta_F\left(\overline{z}^{A}(\mu), \underline{z}^{A}(\mu)\right) \\ - \left(\displaystyle\int_0^{\underline{z}^{A}(\mu)} z\mathrm{d}F(z) + \int_{\overline{z}^{A}(\mu)}^{1} z\mathrm{d}F(z)\right)\end{array}\right]\left[\delta\dfrac{2-\pi}{1-\delta} - c\right]\right)\end{array}\right]}{\left(1 - \delta\pi\Delta_F\left(\overline{z}^{A}(\mu), \underline{z}^{A}(\mu)\right)\right)\left[\delta\dfrac{2-\pi}{1-\delta} - c\right]}. \end{cases} \tag{2}$$

- *The values of* $\overline{\mu}^{A}$ *and* $\underline{\mu}^{A}$ *are, respectively, given as part of the solution to the system of three equations composed of (i) each of the following equations and (2) two equations*

in (2) with $\mu = \overline{\mu}^{\mathtt{A}}$ or $\mu = \underline{\mu}^{\mathtt{A}}$. The thresholds in μ are implicitly specified as

$$\begin{aligned}
\overline{\mu}^{\mathtt{A}} &= \frac{1}{2} + \delta\frac{2-\pi}{1-\delta}\left(\int_0^{\underline{z}^{\mathtt{A}}\left(\overline{\mu}^{\mathtt{A}}\right)} \frac{1}{2} - z\mathrm{d}F_{\overline{\mu}^{\mathtt{A}}}(z) + \int_{\overline{z}^{\mathtt{A}}\left(\overline{\mu}^{\mathtt{A}}\right)}^{1} \frac{1}{2} - z\mathrm{d}F_{\overline{\mu}^{\mathtt{A}}}(z)\right) \\
&\quad + c\left(\int_0^{\underline{z}^{\mathtt{A}}\left(\overline{\mu}^{\mathtt{A}}\right)} z\mathrm{d}F_{\overline{\mu}^{\mathtt{A}}}(z) + \int_{\overline{z}^{\mathtt{A}}\left(\overline{\mu}^{\mathtt{A}}\right)}^{1} z\mathrm{d}F_{\overline{\mu}^{\mathtt{A}}}(z) + \frac{1-\delta}{\delta\pi}\right) \\
\underline{\mu}^{\mathtt{A}} &= \frac{1}{2} + \delta\frac{2-\pi}{1-\delta}\left(\int_0^{\underline{z}^{\mathtt{A}}\left(\underline{\mu}^{\mathtt{A}}\right)} \frac{1}{2} - z\mathrm{d}F_{\underline{\mu}^{\mathtt{A}}}(z) + \int_{\overline{z}^{\mathtt{A}}\left(\underline{\mu}^{\mathtt{A}}\right)}^{1} \frac{1}{2} - z\mathrm{d}F_{\underline{\mu}^{\mathtt{A}}}(z)\right) \\
&\quad - c\left(\int_0^{\underline{z}^{\mathtt{A}}\left(\underline{\mu}^{\mathtt{A}}\right)} 1 - z\mathrm{d}F_{\underline{\mu}^{\mathtt{A}}}(z) + \int_{\overline{z}^{\mathtt{A}}\left(\underline{\mu}^{\mathtt{A}}\right)}^{1} 1 - z\mathrm{d}F_{\underline{\mu}^{\mathtt{A}}}(z) + \frac{1-\delta}{\delta\pi}\right).
\end{aligned}$$

Denote these equilibria by $(\mathtt{A}_1)$ when $\mu < \underline{\mu}^{\mathtt{A}}$ and $(\mathtt{A}_2)$ when $\mu > \overline{\mu}^{\mathtt{A}}$, respectively (as in asymmetric vulnerability). In contrast to the first case (S), these $(\mathtt{A}_1)$ and $(\mathtt{A}_2)$ depict how *expected* (or *inherent*) asymmetry in disaster costs leads to war. To see this, suppose Player 1 is geographically vulnerable to disaster risks, i.e., case $(\mathtt{A}_1)$ where μ is small, so that Player 1 attacks 2 when $D_t = 0$ for sure. Her incentive to fight arises in two steps. First, as in case (S) above, observe that the opportunistic motive for aggression is also at work here. Because we have assumed that μ is small (by case $(\mathtt{A}_1)$), it is likely that Player 2 becomes temporarily advantaged if there is a disaster.

Second, anticipating this, Player 1 faces a fear of future opportunistic aggression and exploitation by the opponent. That is, Player 1 knows that Player 2 will face the temptation to fight in a disaster period if the realized z_t is very small. If Player 1 were to acquiesce to this possibility, she would not win the war because her probability of victory, $p_1^{\mathtt{d}}(z_t) = z_t$, is also very small. As a result, the inherent vulnerability of Player 1 generates a preemptive motive: she has the incentive to attack Player 2 in order to fight under a relatively favorable balance of power.

Figure 3 shows an example when $c = 0.55$, $\delta = 0.5$, $\pi = 0.1$, and $Z_t \sim \text{Beta}(10/9, 10)$ (i.e., $\mu = 0.1$, the solid vertical line in the figure). Observe that the values of the parameters

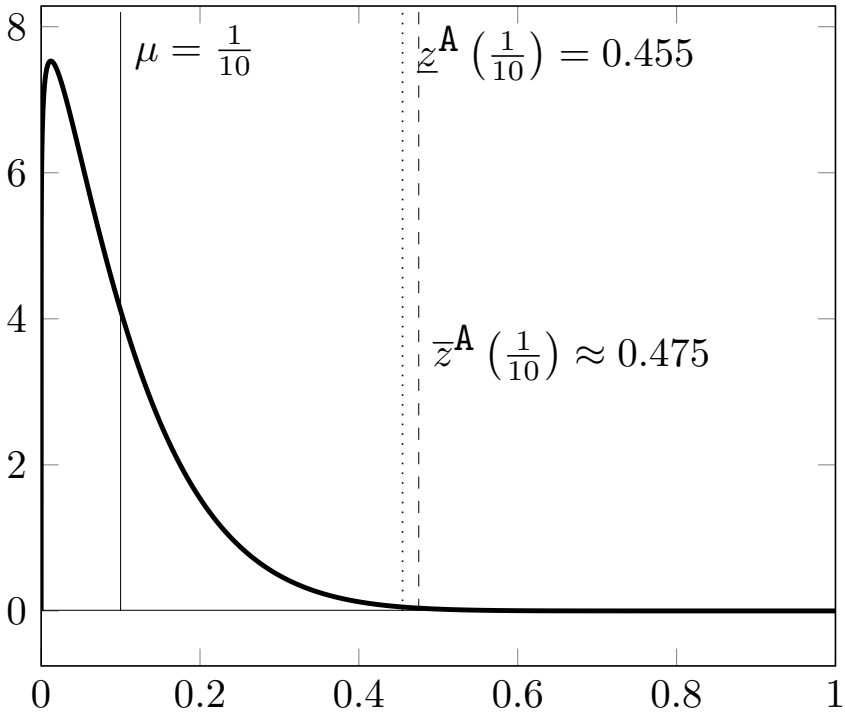


Figure 3: Thresholds in $(\mathtt{A}_1)$ $(c = 0.55, \delta = 0.5, \pi = 0.1, Z_t \sim \text{Beta}(10/9, 10))$

are identical to those in Figure 2 except for the first parameter of the Beta distribution. Compared to the symmetric case ($\mathtt{S}$), the region of realized damage z_t under which war occurs in a disaster period is larger, i.e., $[0, 0.455) \cup (0.475, 1]$. The strategic logic is as follows. First, the inherently vulnerable Player 1 expects that she will likely be disadvantaged and that Player 2 will have an opportunistic motive for conflict once a disaster erupts. Thus, incorporating this risk, Player 1 is willing to fight even when she does not have a military advantage in a disaster period.[12] Second, Player 2 anticipates Player 1's preemptive motive. Therefore, although Player 2 should be militarily advantaged in a disaster period on average (i.e., $\mu = 0.1$), she is also willing to fight even when her actual advantage is not very large.[13]

As the first case above, we can also find examples that suggest the strategic logic of this second, asymmetric vulnerability case. Most notably, as mentioned earlier, Bas and McLean

[12]Notice $p_1^{\mathtt{d}}\left(\overline{z}^{\mathtt{A}}(\mu = 0.1)\right) \approx 0.475 < p_1^{\neg d} = 0.5$. In other words, Player 1 can launch opportunistic aggression when she is slightly *disadvantaged* militarily.

[13]Compare $p_2^{\mathtt{d}}(\mu = 0.1) = 0.9$ and $p_2^{\mathtt{d}}(\underline{z}^{\mathtt{A}}(\mu = 0.1)) = 0.545$.

(2021) find that higher disaster risks are associated with a higher risk of interstate conflict. They obtain this result by restricting attention to years during which neither country in a dyad experienced a disaster. This is consistent with the anticipatory logic, where a conflict erupts even in the absence of an actual disaster event. Moreover, the asymmetric vulnerability case is also compatible with the findings on seasonality and conflict. Landis (2014) shows that, in countries with strong seasonality, civil war is more likely under stable and warm temperatures because fighting is more "strategically viable" under such conditions.[14] An implicit condition for this finding is that political groups *anticipate* a worse strategic environment in a colder season. Without the expectation that military operations today have a higher chance of success than those in a worse climate condition, political leaders do not have a higher incentive to fight today.

3.4 Summary and Comparison

The result below shows that (S), (A_1), and (A_2) exhaust all possible equilibria. As in the above propositions, denote a CDF whose mean is μ by F_μ.

Proposition 3. *Fix all parameters other than μ, and suppose $F_{\overline{\mu}^s}$ and $F_{\underline{\mu}^s}$ satisfy*

$$c < \min \left\{ \begin{array}{l} \dfrac{\frac{1}{2} - \delta \frac{2-\pi}{1-\delta} \left(\int_0^{\underline{z}^s(\overline{\mu}^s)} \frac{1}{2} - z \mathrm{d}F_{\overline{\mu}^s}(z) + \int_{\overline{z}^s(\overline{\mu}^s)}^1 \frac{1}{2} - z \mathrm{d}F_{\overline{\mu}^s}(z) \right)}{\int_0^{\underline{z}^s(\overline{\mu}^s)} z \mathrm{d}F_{\overline{\mu}^s}(z) + \int_{\overline{z}^s(\overline{\mu}^s)}^1 z \mathrm{d}F_{\overline{\mu}^s}(z) + \frac{1-\delta}{\delta\pi}}, \\ \dfrac{\frac{1}{2} + \delta \frac{2-\pi}{1-\delta} \left(\int_0^{\underline{z}^s(\underline{\mu}^s)} \frac{1}{2} - z \mathrm{d}F_{\underline{\mu}^s}(z) + \int_{\overline{z}^s(\underline{\mu}^s)}^1 \frac{1}{2} - z \mathrm{d}F_{\underline{\mu}^s}(z) \right)}{\int_0^{\underline{z}^s(\underline{\mu}^s)} 1 - z \mathrm{d}F_{\underline{\mu}^s}(z) + \int_{\overline{z}^s(\underline{\mu}^s)}^1 1 - z \mathrm{d}F_{\underline{\mu}^s}(z) + \frac{1-\delta}{\delta\pi}} \end{array} \right\}.$$

Then, there exists a unique partition $\left\{0, \underline{\mu}^S, \underline{\mu}^A, \overline{\mu}^A, \overline{\mu}^S, 1\right\}$ such that

[14] See also Koehnlein et al. (2024) for a similar reasoning in the context of violence against civilians.

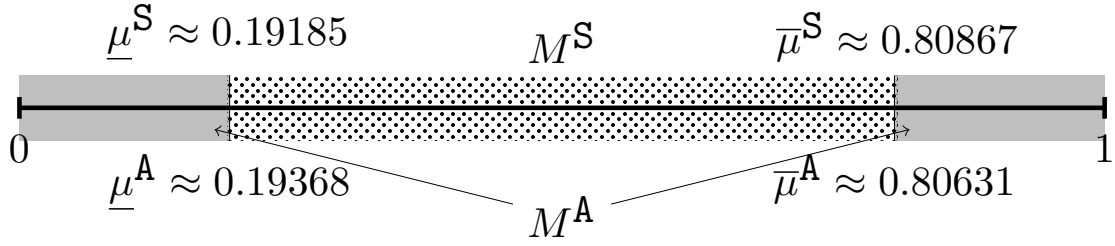


Figure 4: Intervals of μ and equilibrium ($c = 0.15$, $\delta = 0.45$, $\pi = 0.3$, $Z_t \sim \text{Beta}(\alpha, 2)$)

- *when* $\mu < \underline{\mu}^{\mathcal{S}}$, *the unique equilibrium is* $(\mathcal{A}_1)$,
- *when* $\mu \in \left[\underline{\mu}^{\mathcal{S}}, \underline{\mu}^{\mathcal{A}}\right)$, $(\mathcal{S})$ *and* $(\mathcal{A}_1)$ *constitute equilibria,*
- *when* $\mu \in \left[\underline{\mu}^{\mathcal{A}}, \overline{\mu}^{\mathcal{A}}\right]$, *the unique equilibrium is* $(\mathcal{S})$,
- *when* $\mu \in (\overline{\mu}^{\mathcal{A}}, \overline{\mu}^{\mathcal{S}}]$, $(\mathcal{S})$ *and* $(\mathcal{A}_2)$ *constitute equilibria, and*
- *when* $\mu > \overline{\mu}^{\mathcal{S}}$, *the unique equilibrium is* $(\mathcal{A}_2)$,

Figure 4 illustrates the result. Parameters are fixed as $c = 0.15$, $\delta = 0.45$, and $\pi = 0.3$. Z_t follows $\text{Beta}(\alpha, 2)$, and μ takes different values as α varies. In this example, a unique equilibrium exists when μ has a middling value (case $(\mathtt{S})$) and small or large (cases $(\mathtt{A}_1)$ and $(\mathtt{A}_2)$). There are also small regions of equilibrium multiplicity. Approximately speaking, when $\mu \in [0.19185, 0.19386)$, both cases $(\mathtt{S})$ and $(\mathtt{A}_1)$ constitute equilibria. Similarly, when $\mu \in (0.80631, 0.80867]$, cases $(\mathtt{S})$ and $(\mathtt{A}_2)$ are supported as MPE.

We now compare the equilibrium outcomes of the two types of equilibria: $(\mathtt{S})$ on the one hand and $(\mathtt{A}_1)$ and $(\mathtt{A}_2)$ on the other. First, observe that conflict occurs in both disaster and no-disaster periods in cases $(\mathtt{A}_1)$ and $(\mathtt{A}_2)$. In a disaster period, because a mechanism similar to one in case $(\mathtt{S})$ (i.e., opportunistic motive) is at work, conflict erupts with a positive probability. In a no-disaster period, recall that war never occurs in case $(\mathtt{S})$. On the other hand, however, because the vulnerable player expects to be disadvantaged in the future in cases $(\mathtt{A}_1)$ and $(\mathtt{A}_2)$, she preemptively attacks the other with probability one in a no-disaster

period. Thus, in the cases of asymmetric inherent disaster risks, i.e., $(\mathtt{A}_1)$ and $(\mathtt{A}_2)$, war is strictly more likely in the absence of a disaster, whereas it can erupt in both types of periods.

For later use in the following sections, I formally summarize this comparison. Define $Y_t^{\mathtt{game}} = \mathbb{1}\{\text{Conflict occurs at time } t \text{ on the equilibrium path}\}$ and interpret it as a random variable generated by an equilibrium of the model. Further, denote $\mu = \mu^{\mathtt{S}}$ when $\mu \in M^{\mathtt{S}}$ and $\mu = \mu^{\mathtt{A}}$ when $\mu \in M^{\mathtt{A}}$.[15] Then, we obtain the straightforward result below.

Corollary 1. *For any t in each equilibrium, we have*

(i) $\Pr\left(Y_t^{game} = 1 | D_t = 1, \mu^S\right) > \Pr\left(Y_t^{game} = 1 | D_t = 0, \mu^S\right) = 0$, *and*

(ii) $\Pr\left(Y_t^{game} = 1 | D_t = 1, \mu^A\right) < \Pr\left(Y_t^{game} = 1 | D_t = 0, \mu^A\right) = 1$.

Given this corollary, I propose theoretical implications for empirical research to make sense of the seemingly mixed results in the literature.

4 Theoretical Implications for Empirical Research

The theoretical discussion leads to important implications for the empirical literature on climate shocks and armed conflict. The model (Corollary 1) implies that disasters have heterogeneous treatment effects on the risk of armed conflict. That is, in the asymmetric vulnerability case, conflict events should be positively associated with disaster *risks*, rather than actual disaster *events*. Indeed, Bas and McLean (2021) support this implication empirically. To take their findings one step further, this section provides a possible explanation for why the empirical results of climate conflict research appear mixed. First, I show that overlooking the asymmetric vulnerability case of the model (cases $(\mathtt{A}_1)$ and $(\mathtt{A}_2)$) can lead us to underevaluate the effect of natural disasters on conflict risks in the potential outcome framework. Second, based on this discussion, I reinterpret a recent meta-analysis to make sense of the ostensibly mixed empirical results.

[15] For simplicity, assume equilibrium $(\mathtt{A}_1)$ or $(\mathtt{A}_2)$ will be played when $\mu \in M^{\mathtt{S}} \cap M^{\mathtt{A}}$.

4.1 Heterogeneous Effects of Disaster Events

Suppose that we want to employ the potential outcomes framework to study the causal effects of natural disasters and climate anomalies on armed conflict. Let the (binary) potential outcome $Y_{dt}^{\texttt{PO}}(D_{dt}) \in \{0,1\}$ represent the presence ($Y_{dt}^{\texttt{PO}}(D_{dt}) = 1$) and absence ($Y_{dt}^{\texttt{PO}}(D_{dt}) = 0$) of armed conflict at time t in dyad of groups d given treatment D_{dt}.[16] We define the observable treatment D_{dt} in a similar way as in the game-theoretic model. That is, $D_{dt} \in \{0,1\}$ denotes the presence ($D_{dt} = 1$) and absence ($D_{dt} = 0$) of a natural disaster in dyad d in period t, which is independently and identically drawn.

Next, recall that μ is also a *dyadic* concept because it stands for the *relative* (a)symmetry in inherent disaster vulnerabilities of two players. Thus, suppose that we have a measure on the level of such asymmetry for dyad d at time t, denoted by $\mu_{dt} \in [0,1]$. Further, as in the game-theoretic model, suppose that we have two sets $M^{\texttt{S}}, M^{\texttt{A}} \subset [0,1]$ such that (i) rival political groups in dyad d face a symmetric risk of exposure to natural disasters when $\mu_{dt} \in M^{\texttt{S}}$ and (ii) they face an asymmetric risk of exposure when $\mu_{dt} \in M^{\texttt{A}}$.[17] In the potential outcomes framework, this formulation implies that $(Y_{dt}^{\texttt{PO}}(D_{dt})|\mu') \neq (Y_{dt}^{\texttt{PO}}(D_{dt})|\mu'')$, where $\mu' \in M^{\texttt{S}}, \mu'' \in M^{\texttt{A}}$. Because we focus on the role of political groups' relative vulnerabilities to natural disasters (measured by μ_{dt}), our quantity of interest here is the conditional average treatment effect (CATE), defined as $\tau(\mu) \equiv \mathbb{E}[Y_{dt}^{\texttt{PO}}(1) - Y_{dt}^{\texttt{PO}}(0)|\mu_{dt} = \mu]$.

To derive implications for empirical research, suppose that we have country-year data of conflict events in dyad d at time t. Denote the presence and absence of conflict by $Y_{dt}^{\texttt{data}} = 1$ and $Y_{dt}^{\texttt{data}} = 0$, respectively. Further, let us tentatively assume that the game-theoretic model captures the true data-generating process.

Assumption 2. We can interpret $\underbrace{Y_{dt}^{\texttt{data}}}_{\text{Data}} = \underbrace{Y_t^{\texttt{game}}}_{\text{Model}}$ and $\underbrace{D_{dt}}_{\text{Data}} = \underbrace{D_t}_{\text{Model}}$.

[16] One can think of d as a pair of rival political groups or sovereign states. Because the potential outcome of conflict for observation dt is solely determined by D_{dt}, here we are implicitly establishing the stable unit treatment value assumption (SUTVA). For example, we assume that a natural disaster at time t affecting a dyad of groups d does not affect the risk of conflict in a neighboring dyad d'.

[17] For simplicity, assume that $M^{\texttt{S}}$ and $M^{\texttt{A}}$ are mutually disjoint.

In other words, we interpret real-world conflict data as products of equilibrium outcomes in the game of asymmetric exposure to extreme weather events. Then, given that $Y_{dt}^{\texttt{data}} = Y_{dt}^{\texttt{PO}}(1)$ when $D_t = 1$, we obtain the following.

Corollary 2. *Assumption 2 and Corollary 1 yield:*

(i) $\Pr\left(Y_{dt}^{PO}(1) = 1 | \mu_{dt} = \mu^S\right) > \Pr\left(Y_{dt}^{PO}(0) = 1 | \mu_{dt} = \mu^S\right)$ *and*

(ii) $\Pr\left(Y_{dt}^{PO}(1) = 1 | \mu_{dt} = \mu^A\right) < \Pr\left(Y_{dt}^{PO}(0) = 1 | \mu_{dt} = \mu^A\right).$

In words, the result represents the two types of equilibrium outcomes: (i) when two rival political groups face a symmetric risk of exposure to disasters, war is more likely to erupt in a given period t when a disaster has occurred in the same period (i.e., $D_{dt} = 1$); (ii) when they face an asymmetric risk of exposure, conflict is more likely when there is no disaster in period t (i.e., $D_{dt} = 0$).

This result drawn from the game-theoretic model leads to another, more specific implication. Suppose $\mu_{dt} = \mu^{\texttt{S}}$. Rearranging Part (i) of Corollary 2 immediately yields

$$\begin{aligned}
\Pr\left(Y_{dt}^{\texttt{PO}}(1) = 1 \middle| \mu^{\texttt{S}}\right) - \Pr\left(Y_{dt}^{\texttt{PO}}(0) = 1 \middle| \mu^{\texttt{S}}\right) &= 1 \cdot \Pr\left(Y_{dt}^{\texttt{PO}}(1) = 1 \middle| \mu^{\texttt{S}}\right) + 0 \cdot \Pr\left(Y_{dt}^{\texttt{PO}}(1) = 0 \middle| \mu^{\texttt{S}}\right) \\
&\quad -1 \cdot \Pr\left(Y_{dt}^{\texttt{PO}}(0) = 1 \middle| \mu^{\texttt{S}}\right) + 0 \cdot \Pr\left(Y_{dt}^{\texttt{PO}}(0) = 0 \middle| \mu^{\texttt{S}}\right) \\
&= \mathbb{E}\left[Y_{dt}^{\texttt{PO}}(1) \middle| \mu^{\texttt{S}}\right] - \mathbb{E}\left[Y_{dt}^{\texttt{PO}}(0) \middle| \mu^{\texttt{S}}\right] \\
&= \tau\left(\mu^{\texttt{S}}\right) > 0.
\end{aligned}$$

Similarly, from Part (ii) of Corollary 2, we obtain $\Pr\left(Y_{dt}^{\texttt{PO}}(1) = 1 | \mu^{\texttt{A}}\right) - \Pr\left(Y_{dt}^{\texttt{PO}}(0) = 1 | \mu^{\texttt{A}}\right) = \tau\left(\mu^{\texttt{A}}\right) < 0$. Implication 1 below summarizes the results.

Implication 1. Corollary 2 implies $\tau\left(\mu^{\texttt{S}}\right) > 0 > \tau\left(\mu^{\texttt{A}}\right)$.

It is important to note that, even if researchers rely on CATE rather than ATE, the potential outcome approach itself could be problematic in this case. To see this, suppose

that belligerent groups in dyad d face an asymmetric risk of exposure to disasters (i.e., $\mu_{dt} = \mu^{\mathtt{A}}$). Then, Implication 1 states that the effect of a disaster on conflict onset in the sense of CATE is negative, $\tau(\mu^{\mathtt{A}}) < 0$, under Corollary 2.

However, although the CATE is negative, asymmetric exposure to disaster risks triggers a war in the game-theoretic model: Remarks 1 and 2 show that the possibility of disasters and variation in Z_t are necessary conditions for a conflict to erupt on the equilibrium path. Therefore, if asymmetric exposure to disasters and climate anomalies generates an *anticipation* of temporary military power shifts in the real world (as empirically tested by Bas and McLean, 2021), overlooking the strategic dynamic of the second equilibrium might induce the researcher to underevaluate their true effect on the propensity of war.

4.2 Relative Vulnerabilities and the Average Treatment Effect

Given the above discussion on CATE, let us move on to average treatment effect (ATE), defined as $\tau \equiv \mathbb{E}[Y_{dt}^{\mathtt{PO}}(1) - Y_{dt}^{\mathtt{PO}}(0)]$, which is the most common quantity of interest in the empirical literature. Below, I show that we are more likely to obtain a negative estimate for the ATE of natural disasters on conflict when the number of observations involving asymmetric dyads of belligerent groups in the sample is sufficiently large.

To this end, we interpret the parameter μ as a random variable that represents the level of asymmetry in disaster risks for two relevant groups, and denote it by $\mu^{\mathtt{rdm}}$.[18] Assume that $\mu^{\mathtt{rdm}}$ is independent of the other random variables in the model, D_t and Z_t. Moreover, as in Proposition 3, suppose there are five regions over $[0, 1]$ that determine which strategy profiles constitute equilibria. Then, denote $q^{\mathtt{S}} \equiv \Pr(\mu^{\mathtt{rdm}} \in M^{\mathtt{S}} \backslash M^{\mathtt{A}})$ and $q^{\mathtt{A}} \equiv \Pr(\mu^{\mathtt{rdm}} \in M^{\mathtt{A}} \backslash M^{\mathtt{S}})$ so that $\Pr(\mu^{\mathtt{rdm}} \in M^{\mathtt{S}} \cap M^{\mathtt{A}}) = 1 - q^{\mathtt{S}} - q^{\mathtt{A}}$. Recall that there are multiple equilibria when $\mu^{\mathtt{rdm}} \in M^{\mathtt{S}} \cap M^{\mathtt{A}}$. Denote the strategy profile realized as an equilibrium by $(\mathtt{E})$, and let

[18] Recall that μ is an exogenous parameter in the game. However, it is also natural to think of the asymmetry in inherent disaster risks between two rival political groups as randomly distributed among dyads. Here, we take this interpretation to show the connection between the theory and data.

$\tilde{q} \equiv \Pr\left((\mathtt{E}) = (\mathtt{S}) | \mu^{\mathtt{rdm}} \in M^{\mathtt{S}} \cap M^{\mathtt{A}}\right)$ so that $\Pr\left((\mathtt{E}) = (\mathtt{A}) | \mu^{\mathtt{rdm}} \in M^{\mathtt{S}} \cap M^{\mathtt{A}}\right) = 1 - \tilde{q}$. Then, by the law of iterated expectations, it is easy to see that the ATE is given by

$$\begin{aligned}
\tau &= \mathbb{E}_{\mu^{\mathtt{rdm}}}\left[\underbrace{\mathbb{E}_{D_{dt}}\left[Y_{dt}^{\mathtt{PO}}(1) - Y_{dt}^{\mathtt{PO}}(0) \middle| \mu^{\mathtt{rdm}} = \mu\right]}_{\text{CATE } \tau(\mu)}\right] \\
&= \tau\left(\mu^{\mathtt{S}}\right)\left[q^{\mathtt{S}} + \underbrace{\left(1 - q^{\mathtt{S}} - q^{\mathtt{A}}\right)\tilde{q}}_{\Pr(\mu^{\mathtt{rdm}} \in M^{\mathtt{S}} \cap M^{\mathtt{A}}, (\mathtt{E}) = (\mathtt{S}))}\right] + \tau\left(\mu^{\mathtt{A}}\right)\left[q^{\mathtt{A}} + \underbrace{\left(1 - q^{\mathtt{S}} - q^{\mathtt{A}}\right)(1 - \tilde{q})}_{\Pr(\mu^{\mathtt{rdm}} \in M^{\mathtt{S}} \cap M^{\mathtt{A}}, (\mathtt{E}) = (\mathtt{A}))}\right].
\end{aligned}$$

This leads to the following theoretical result.

Corollary 3. *If* $-\dfrac{\tau\left(\mu^{S}\right)}{\tau\left(\mu^{A}\right)} < \dfrac{q^{A}}{1 - q^{A}}$, *then* $\tau < 0$.

This result is intuitive. It states that, assuming that μ is now random, the ATE of natural disasters on conflict in the model becomes negative if the asymmetric case is more likely to arise as an equilibrium. This is because $\tau\left(\mu^{\mathtt{A}}\right)$ is negative and a higher $q^{\mathtt{A}}$ allocates a larger weight on the negative CATE.

In a similar manner, now consider a simple finite-sample decomposition. Consider a study k with n_k observations. As above, write conflict and disaster event for observation i as binary indicators Y_{ik} and D_{ik}, respectively. Suppose we also have a binary measure of asymmetric vulnerability, $A_{ik} \in \{0, 1\}$, where $A_{ik} = 1$ and $A_{ik} = 0$ respectively mean that observation i of study k involves groups with asymmetric and symmetric vulnerability. Also, let $q_k^{\mathtt{A}} = \Pr\left(A_{ik} = 1\right)$ for all i in study k. Then, define subgroup difference-in-means estimators:

$$\begin{aligned}
\hat{\tau}_k^{\mathtt{S}} &= \frac{\sum_{i=1}^{n_k} D_{ik}\left(1 - A_{ik}\right) Y_{ik}}{\sum_{i=1}^{n_k} D_{ik}\left(1 - A_{ik}\right)} - \frac{\sum_{i=1}^{n_k}\left(1 - D_{ik}\right)\left(1 - A_{ik}\right) Y_{ik}}{\sum_{i=1}^{n_k}\left(1 - D_{ik}\right)\left(1 - A_{ik}\right)} \\
\hat{\tau}_k^{\mathtt{A}} &= \frac{\sum_{i=1}^{n_k} D_{ik} A_{ik} Y_{ik}}{\sum_{i=1}^{n_k} D_{ik} A_{ik}} - \frac{\sum_{i=1}^{n_k}\left(1 - D_{ik}\right) A_{ik} Y_{ik}}{\sum_{i=1}^{n_k}\left(1 - D_{ik}\right) A_{ik}}.
\end{aligned}$$

As the number of observations in study k involving groups with symmetric/asymmetric

vulnerabilities, denote $n_k^{\mathtt{S}} = \sum_{i=1}^{n_k} (1 - A_{ik})$ and $n_k^{\mathtt{A}} = \sum_{i=1}^{n_k} A_{ik} = n_k - n_k^{\mathtt{S}}$. Then, letting $\hat{q}_k^{\mathtt{A}} = n_k^{\mathtt{A}}/n_k$, we are interested in $\hat{\tau}_k = \hat{q}_k^{\mathtt{A}}\hat{\tau}_k^{\mathtt{A}} + (1 - \hat{q}_k^{\mathtt{A}})\,\hat{\tau}_k^{\mathtt{S}}$. Hence, if we have positive $\hat{\tau}_k^{\mathtt{S}}$ and negative $\hat{\tau}_k^{\mathtt{A}}$, it is easy to see that the quantity of interest $\hat{\tau}_k$ becomes negative if and only if

$$\frac{\hat{\tau}_k^{\mathtt{S}}}{-\hat{\tau}_k^{\mathtt{A}}} < \frac{n_k^{\mathtt{A}}}{n_k^{\mathtt{S}}}.$$

Implication 2. If the number of observations involving dyads with asymmetric vulnerabilities is sufficiently large in the sample, then an estimate for the ATE of natural disasters on conflict is more likely to be small.

Recall that, if natural disasters generate a random and large negative shock to political groups' military capabilities, they can trigger conflict in *both* symmetric and asymmetric dyads due to either the opportunistic or preemptive motive. However, if an empirical study contains disproportionately more dyads of groups with asymmetric vulnerabilities to disasters than symmetric ones in its sample, then such a study is more likely to understate the conflict-inducing effect of natural disasters.

4.3 Reinterpreting Existing Empirical Studies

We are now ready for a simple exercise that connects the theory to empirics: I apply the above implication to reinterpret the existing studies in a recent meta-analysis by Burke et al. (2024). Burke et al. (2024) collect research articles that examine the effect of climate conditions on the risk of intergroup conflict. They reanalyze these studies using a common empirical model and present estimates of the effect. As a coarse measure of relative vulnerabilities to future disasters, I use the group types in the datasets used in the studies.

Assumption 3. An intrastate conflict involving the government as one of the belligerents tends to be between groups with asymmetric vulnerabilities rather than a conflict involving only non-state actors and one involving only government forces.

In other words, given that two groups are engaged in a conflict, we interpret that they face symmetric vulnerabilities to disasters if the types of the involved parties are similar (e.g., social conflicts between non-state actors and coups among different factions in the government). Conversely, I assume that they tend to have asymmetric vulnerabilities when one side is part of the government forces and the other is not (e.g., civil wars). It is natural to suppose that, on average, government forces tend to be more resilient to disasters than rebel groups are. Though an imperfect measure, this assumption is consistent with some existing studies (e.g., Tominaga and Lee, 2021; Walch, 2018).[19]

We say study k in Burke et al. (2024) is more likely to contain conflict events between asymmetric vulnerabilities when k uses a dataset of civil conflicts (e.g., the UCDP/PRIO dataset, Gleditsch et al., 2002), and I denote this case as $k \in S^{\mathtt{A}}$. On the other hand, denote $k \in S^{\mathtt{S}}$ when study k relies on a dataset of non-state social conflicts, riots, or coups to describe events that are more likely to be between groups with symmetric vulnerabilities. Some datasets contain both types of events, such as Armed Conflict Location and Event Data (ACLED, Raleigh et al., 2023), in which case I denote such indeterminate studies by $k \in S^{\mathtt{I}}$. I exclude five studies from the following exercise because they are interested in nonviolent events, such as regime change, democratization, and nonviolent protests, which are outside of the scope of my theory. This categorization results in 12 studies in $S^{\mathtt{S}}$, 14 in $S^{\mathtt{A}}$, and 16 in $S^{\mathtt{I}}$. Table 2 in Appendix D shows the breakdown of the classification.

To assess whether the asymmetric-vulnerability theory is useful to reinterpret the seemingly mixed empirical findings, we are interested in the sign of the estimated effects of climate events on the risk of conflict. Below, I present an explicit empirical implication, analogous to Corollary 1 and Implication 2, about how often we should see a positive or negative

[19] One possible problem is selection effects: rebel groups that have enough material capabilities to fight government forces may also be resilient. However, relatively weak rebel groups can still continue armed struggles with the (more powerful) government in some environments. For example, conflicts in locations distant from the capital tend to last longer because government control in those regions tends to be weaker (Buhaug et al., 2009). Moreover, even weaker groups may be able to endogenously render their armed struggle last longer by adopting guerrilla tactics (Qiu, 2022).

estimated effect of climate events, depending on the type of each study in a group of empirical studies. Thus, we are interested in the conditional probabilities $\Pr(\hat{\tau}_k > 0 | k \in S^{\mathtt{S}})$ and $\Pr(\hat{\tau}_k > 0 | k \in S^{\mathtt{A}})$.

Suppose we can express an unbiased estimator of the ATE in study k as $\hat{\tau}_k = \tau_k + \varepsilon_k$, where τ_k is the true ATE of study k and the random variable ε_k is continuous and symmetric around zero. We use $q_k^{\mathtt{A}}$ and $q_k^{\mathtt{S}} = 1 - q_k^{\mathtt{A}}$, defined above, as the ratio of observations involving asymmetric group types among all observations in study k. By the discussion above, we have $\tau_k = q_k^{\mathtt{A}} \tau_k(\mu^{\mathtt{A}}) + (1 - q_k^{\mathtt{A}}) \tau_k(\mu^{\mathtt{S}})$. Then, suppose that $q_k^{\mathtt{A}}$ itself is a random variable over $[0, 1]$ with CDFs $G^{\mathtt{S}}(q) = \Pr(q_k^{\mathtt{A}} \leq q | k \in S^{\mathtt{S}})$ and $G^{\mathtt{A}}(q) = \Pr(q_k^{\mathtt{A}} \leq q | k \in S^{\mathtt{A}})$. Moreover, assume that the type of study k affects $\hat{\tau}_k$ only through the ratio of observations that involve groups with asymmetric vulnerabilities $q_k^{\mathtt{A}}$, so that $\Pr(\hat{\tau}_k > 0 | q_k^{\mathtt{A}} = q, k \in S^{\mathtt{S}}) = \Pr(\hat{\tau}_k > 0 | q_k^{\mathtt{A}} = q, k \in S^{\mathtt{A}}) = \Pr(\hat{\tau}_k > 0 | q_k^{\mathtt{A}} = q)$. Finally, analogous to Assumption 3, suppose that $G^{\mathtt{A}}$ is first-order stochastic dominant (FOSD) over $G^{\mathtt{S}}$:

Assumption 3'. $G^{A}(q) \leq G^{S}(q)$ for all $q \in (0, 1)$, and there exists q' such that $G^{\mathtt{A}}(q') < G^{\mathtt{S}}(q')$.

Note that this assumption and Assumption 3 are not as stringent as they might look. In reality, for example, some rebel groups with mobile military capabilities may be more resilient to disaster damages compared to state forces with fixed military facilities. However, Assumption 3' allows a case in which, for a pair of two specific studies $k' \in S^{\mathtt{S}}$ and $k'' \in S^{\mathtt{A}}$, we have $q_{k'}^{\mathtt{A}} > q_{k''}^{\mathtt{A}}$. The assumption does *not* require that studies examining conflicts between different group types always feature asymmetric vulnerabilities. It merely supposes the ratio of dyads with asymmetric vulnerabilities *tends* to be higher in a study focusing on conflicts between the government and rebel groups.

This formulation leads to the following prediction about the estimated effect of climate events on armed conflict in empirical studies.

	Symmetric: Predicts $\hat{\tau}_k > 0$	Asymmetric: Predicts $\hat{\tau}_k < 0$	
Positive effect	**9**	**6**	15
	e.g., Kim (2016)	e.g., Levy et al. (2005)	
	Temperature and coup	Drought and civil war	
Negative effect	**3**	**8**	11
	e.g., Helman et al. (2020)	e.g., Salehyan and Hendrix (2014)	
	Temperature and non-state conflict	Drought and civil war	
	12	14	26

Table 1: Reinterpreting Burke et al. (2024)

Proposition 4. *If* $\tau_k(\mu^S) > \tau_k(\mu^A)$ *for all* k*, we have*

$$\Pr\left(\hat{\tau}_k > 0 \middle| k \in S^S\right) > \Pr\left(\hat{\tau}_k > 0 \middle| k \in S^A\right).$$

Using this proposition and studies in Burke et al. (2024), I illustrate how the theory helps us reinterpret empirical results. As shown in Table 1, nine of 12 studies in Burke et al. (2024) categorized as part of $S^{\mathtt{S}}$ show a positive estimated coefficient. On the other hand, six of 14 studies in $S^{\mathtt{A}}$ suggest a positive effect of climate events on conflict. Let S_+ denote these studies with $\hat{\tau}_k > 0$. Note that the inequality in Proposition 4 is equivalent to $\frac{\Pr\left(\hat{\tau}_k>0, k\in S^{\mathtt{S}}\right)}{\Pr(k\in S^{\mathtt{S}})} > \frac{\Pr\left(\hat{\tau}_k>0, k\in S^{\mathtt{A}}\right)}{\Pr(k\in S^{\mathtt{A}})}$. To apply this to Burke et al. (2024), we use the following proxies for each case of relative vulnerability $\mathtt{R} \in \{\mathtt{S}, \mathtt{A}\}$: $\frac{|S^{\mathtt{R}} \cap S_+|}{|S^{\mathtt{S}} \cup S^{\mathtt{A}}|}$ for $\Pr\left(\hat{\tau}_k > 0, k \in S^{\mathtt{R}}\right)$ and $\frac{|S^{\mathtt{R}}|}{|S^{\mathtt{S}} \cup S^{\mathtt{A}}|}$ for $\Pr\left(k \in S^{\mathtt{R}}\right)$, where $|S|$ denotes the number of elements of a finite set S. Then, we obtain the ratio $\frac{|S^{\mathtt{R}} \cap S_+|}{|S^{\mathtt{R}}|} \equiv \rho^{\mathtt{R}}$ in studies examined in Burke et al. (2024) to express the probability $\Pr\left(\hat{\tau}_k > 0 | k \in S^{\mathtt{R}}\right)$.

Implication 3. Proposition 4 implies $\rho^{\mathtt{S}} > \rho^{\mathtt{A}}$ in Burke et al. (2024).

To illustrate how unlikely it should be to see the two-by-two relationship in Table 1 if different relative vulnerabilities do not play any role, I apply Fisher's exact test. Namely, I consider Table 1 as the record of my theory's performance in a hypothetical experiment to predict the estimated effect of the climate variable in studies examined by Burke et al. (2024). Given the prediction $\Pr\left(\hat{\tau}_k > 0 | k \in S^{\mathtt{S}}\right) > \Pr\left(\hat{\tau}_k > 0 | k \in S^{\mathtt{A}}\right)$ in Proposition 4 (or,

analogously, $\rho^{\mathtt{S}} > \rho^{\mathtt{A}}$ in Implication 3), Fisher's exact test (one-sided) yields a Fisher's p-value of 0.1042.[20] Thus, although not significant at a level widely used in the social sciences literature, if there were no true association between the symmetric- and asymmetric-vulnerability cases, we would see this or a more favorable result roughly 10 percent of the time. A two-sided test leads to a Fisher's p-value of 0.1302. In sum, while the results of this simple exercise are only suggestive, the asymmetric-vulnerability theory provides a novel interpretation for why empirical assessments of the effect of extreme climate events on conflict are likely to be inconclusive.

A caveat is that there are 16 "indeterminate" studies $S^{\mathtt{I}}$, where the conflict data of study k in Burke et al. (2024) contains conflict events involving both symmetric and asymmetric group types. For transparency, I examine how these cases affect the simple empirical exercise by reporting the upper and lower bounds of $\rho^{\mathtt{S}} - \rho^{\mathtt{A}}$. As shown in Appendix D, only one study in $S^{\mathtt{I}}$ (Guardado and Pennings, 2025, GP henceforth) presents a negative estimated effect of climate events on conflict, whereas the other 15 indeterminate studies report a positive effect. The best-case scenario that fits the theory is that $\{\text{GP}\} \in S^{\mathtt{A}}$ and $k \in S^{\mathtt{S}}$ for all $k \in S^{\mathtt{I}} \backslash \{\text{GP}\}$. On the other hand, the worst-case scenario is one in which only GP is in $S^{\mathtt{S}}$ and the other 15 studies are in $S^{\mathtt{A}}$. These two extreme cases yield $\rho^{\mathtt{S}} - \rho^{\mathtt{A}} \in [-0.03183, 0.48889]$. While this interval includes negative values, the sign of the difference flips into negative for only one, the least favorable case above.

4.4 Reconsidering the Possible Pacifying Effect of Disasters

Here, I reconsider a specific existing empirical result included in Burke et al. (2024) from a theoretical perspective. Namely, I discuss the results presented by Salehyan and Hendrix

[20] More precisely, the null in Fisher's exact test assumes that the marginal totals are fixed and the two distributions are identical, which is $\Pr(\hat{\tau}_k > 0 | k \in S^{\mathtt{S}}) = \Pr(\hat{\tau}_k > 0 | k \in S^{\mathtt{A}})$ in this context. A "one-sided" p-value here means that we sum up the probabilities that, assuming $\Pr(\hat{\tau}_k > 0 | k \in S^{\mathtt{S}}) = \Pr(\hat{\tau}_k > 0 | k \in S^{\mathtt{A}})$, we obtain (i) the allocation of the number of studies given in the table or (ii) more favorable allocations that match the prediction better.

(2014), who find a possible pacifying effect of droughts, and I point out that their findings could be due to the preemptive motive for conflict caused by future disaster risks. To be clear, the discussion below is not aimed at falsifying Salehyan and Hendrix (2014). Rather, both Salehyan and Hendrix (2014) and the present paper consider the military implications of climate disasters. With this caveat, I argue that my theory and theirs generate *observationally equivalent* predictions. That is, some empirical findings on the negative causal effects of climate disasters on conflict could be driven by a particular form of their *positive* effect, namely, the preemptive mechanism in the asymmetric cases ($\mathtt{A}_1$) and ($\mathtt{A}_2$) of the model.

Salehyan and Hendrix (2014) argue that water scarcity can decrease armed conflict. Droughts harm agricultural production in affected areas and reduce economic resources. As a result, water scarcity impairs the mobilization capabilities of rival political groups, and thus, organized armed conflict should decline under such conditions. Utilizing data on water scarcity and armed conflict, Salehyan and Hendrix (2014) find that water abundance positively correlates with conflict.[21]

Their interpretation of the correlation might be problematic. To see this, observe the implicit assumption in the logic of Salehyan and Hendrix (2014) that water scarcity affects groups' mobilization capabilities symmetrically. Namely, conflict becomes a less attractive option under water scarcity because, in their theory, *both* sides of a given dispute face a lower mobilization capability. However, if one drought happens to cause water scarcity in a particular region, such an (implicit) assumption does not necessarily remain plausible. Rather, if water scarcity is more severe in one region than in others, then only one side of the warring parties can lose its power to mobilize. In this case, the rival group that has incurred less severe water scarcity retains its mobilization capability. As the opportunistic motivation in the formal model suggests, such asymmetry in power can generate an incentive for rival political groups to fight, rather than deescalate.

[21] See also Ide (2023) for a similar argument, who finds that ongoing conflicts can deescalate when a disaster affects both sides of the dispute severely. Nemeth and Lai (2022) also argue that belligerent groups are more likely to bargain when a disaster has caused similar levels of damage to each of them.

Given the demanding implicit assumption on the symmetry in disaster damages, the findings of Salehyan and Hendrix (2014) might be a product of preemptive motives arising from the asymmetric vulnerability equilibrium ($\mathtt{A}_1$) or ($\mathtt{A}_2$). That is, the equilibrium suggests that conflict is strictly more likely in periods without water scarcity because the inherently vulnerable side is willing to gamble on fighting in the *absence* of an actual climate shock. We thus have contrasting theoretical reasoning. Salehyan and Hendrix (2014) see a climate shock as a potential source of political stability due to deteriorated mobilization capability. The present paper, on the other hand, considers the risk of such shocks as a source of instability due to the preemptive motive. However, observe that *both* the mobilization-capability logic and the preemptive-motive logic predict a negative correlation between disaster events and conflict events. Thus, these two opposing logics lead to the same empirical pattern.

Implication 4. Despite their opposing (destabilizing versus pacifying) conclusions, the preemptive-attack logic and the mobilization-capability logic generate observationally equivalent predictions.

One can also situate other studies that seem to contradict Salehyan and Hendrix (2014) in the context of the asymmetric-vulnerability theory. For example, Fjelde and von Uexkull (2012) find that negative rainfall anomalies are associated with more communal conflicts between non-state actors in sub-Saharan Africa. Due to their focus on non-state conflict, the paper is categorized in $S^{\mathtt{S}}$, according to the criterion above. Because conflict datasets involving only non-state conflict events are more likely to contain groups with symmetric vulnerabilities, the finding of Fjelde and von Uexkull (2012) that droughts and communal conflicts are positively correlated is consistent with the theory.

Moreover, Bell and Keys (2018) argue that the effect of drought on conflict in sub-Saharan Africa is conditional on other factors related to conflict risks, such as social vulnerability (e.g., inadequate food supply), equal distribution of or access to emergency resources (e.g., ethnic exclusion), and state capacity (e.g., low urbanization). For instance, they find a significant

positive association between drought and conflict contingent on low ethnic exclusion. If inclusive distribution of resources among different ethnic groups implies that those groups have relatively symmetric vulnerabilities to extreme weather events ($\mu \in M^{\mathtt{S}}$ in the model), this finding exemplifies Corollary 1. Hence, it is also consistent with the asymmetric-vulnerability theory.

In sum, some of the seemingly contradictory empirical results can be driven by one of the two heterogeneous effects of disasters conditional on the level of (a)symmetry in disaster risks. We may be able to make sense of them by taking theory and the data-generating process behind them more seriously.

5 Conclusion

This paper has shown that a single process of asymmetric exposure to climate anomalies and other natural disasters generates two qualitatively distinct equilibrium strategies. The key parameter is the asymmetry in the inherent vulnerability to extreme weather events. It determines the source of conflict in the equilibrium path. In the first case, conflict erupts as a consequence of an actual disaster that causes a rapid shift in the balance of power. The sole trigger of a conflict is an opportunistic motive of the temporarily advantaged side. In the second, war is more likely before a disaster occurs because of the anticipated asymmetry of its costs. While the opportunistic and preemptive motives coexist in this case, the latter overwhelms the former. Put differently, although disasters trigger conflict in both cases either through their actual occurrence or their future risks, the probability of conflict onset in the asymmetric vulnerability case is strictly larger in the absence of a disaster than right after it.

This theoretical result yields a novel theoretical reinterpretation of why the results of empirical research on climate conflict are inconclusive. Although it does not directly dispute specific empirical results, the game-theoretic model implies that we might be understating the

causal effect of disasters and extreme weather events on armed conflict. Specifically, in the asymmetric vulnerability case, the actual occurrence of a natural disaster and conflict onset should be negatively correlated, but those conflicts are also caused by disaster risks. Therefore, if extreme weather events have multiple effects, including the one studied in this paper (randomness and asymmetry in relative vulnerabilities to future disasters), the seemingly mixed results may be mixed only ostensibly. Namely, it is possible that the conflict-inducing effects of natural disasters are larger than shown in empirical studies if they overlook the heterogeneous impacts.

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Appendix for:

Natural Disasters, Asymmetric Vulnerabilities, and War

Hiroto Sawada

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A Proofs of the Statements in the Main Text

A.1 Proof of Remark 1

Proof. We want to show that (i) a strategy profile in which both players never fight constitutes an SPE and (ii) it is the unique equilibrium. First, consider an arbitrary subgame, and suppose that one player is following a strategy in which she never attacks the opponent at any decision node. Let $\mathcal{V}^0$ and $\mathcal{W}^0$ denote the other player's continuation payoff and expected war payoff, respectively. Then, we have

$$\begin{aligned}
\mathcal{V}^0 &= \theta + \delta \mathcal{V}^0 \\
\mathcal{W}^0 &= \frac{1}{2}\left[2\theta(1-c) + \delta \sum_{t=1}^{\infty} \delta^{t-1} \cdot 2\theta\right] + \frac{1}{2} \cdot 0 \\
&= \theta\left[\frac{1}{1-\delta} - c\right].
\end{aligned}$$

Since solving the first equation for $\mathcal{V}^0$ yields $\mathcal{V}^0 = \theta/(1-\delta)$, we obtain $\mathcal{V}^0 > \mathcal{W}^0$ for any $c \in (0,1)$. Because any deviation (attacking the opponent) ends the game, we do not need to check off-path strategies. Hence, the strategy profile in which neither player ever fights constitutes an SPE.

The above equilibrium is unique under Assumption 1. To see this, suppose to the contrary that there exists another SPE. If so, because such an equilibrium is different from the one shown above in which neither attacks the other at any history, it implies that war must occur in some history. Assumption 1 rules out strategy profiles in which both players attack in the same period. Thus, it leaves a strategy profile where one player unilaterally attacks the other at some history t'. Consider the attacker's incentive at such period t'. For an arbitrary continuation game, a one-step deviation of not attacking at period t' would yield a path of play on which peace is maintained for $\tau \geq 1$ periods and war erupts in the $(\tau+1)$th period (unless $\tau = \infty$). Such deviation would generate a continuation payoff $\sum_{t=1}^{\tau} \delta^{t-1}\theta + \delta^{\tau}\mathcal{W}^0$. In

sum, the incentive of the attacker at period t' has to satisfy

$$\begin{aligned} \underbrace{\mathcal{W}^0}_{\text{Attack at } t'} &\geq \underbrace{\frac{1-\delta^\tau}{1-\delta}\theta + \delta^\tau \mathcal{W}^0}_{\text{Deviation}} \\ -c &\geq 0, \end{aligned}$$

which never holds by $c \in (0,1)$. Therefore, any potential attacker has the incentive to deviate to sustain peace in any strategy profile where war occurs at some history t'. This result contradicts the assumption that there is another SPE, which proves that no other equilibrium is possible under Assumption 1. □

A.2 Proof of Remark 2

Proof. As in Remark 1, we want to show that the completely peaceful strategy profile is the only SPE. To show this, first define players' continuation payoff from the strategy profile:

$$\begin{aligned} \mathcal{V}^{\mathtt{sym}} &\equiv (1-\pi)\left(\theta + \delta \mathcal{V}^{\mathtt{sym}}\right) + \pi\left(\frac{1}{2}\theta + \delta \mathcal{V}^{\mathtt{sym}}\right) \\ \mathcal{V}^{\mathtt{sym}} &= \theta \cdot \frac{1-\pi/2}{1-\delta}. \end{aligned}$$

Next, let $\mathcal{W}^{\mathtt{d},\mathtt{sym}}$ and $\mathcal{W}^{\neg\mathtt{d},\mathtt{sym}}$ be the expected war payoff when a war occurs in a disaster and no-disaster periods, respectively.

$$\begin{aligned} \mathcal{W}^{\mathtt{d},\mathtt{sym}} &= \frac{1}{2}\left(\theta(1-c) + \delta\overline{\mathcal{V}}\right) \\ \mathcal{W}^{\neg\mathtt{d},\mathtt{sym}} &= \frac{1}{2}\left(2\theta(1-c) + \delta\overline{\mathcal{V}}\right). \end{aligned}$$

Not attacking is the best response in disaster and no-disaster periods if $\frac{1}{2}\theta + \delta\mathcal{V}^{\mathtt{sym}} > \mathcal{W}^{\mathtt{d},\mathtt{sym}}$ and $\theta + \delta\mathcal{V}^{\mathtt{sym}} > \mathcal{W}^{\neg\mathtt{d},\mathtt{sym}}$, respectively. Simple computation reduces both inequalities to $c > 0$, which always holds. Because not attacking is the best response to the other player not

attacking in any subgame, the strategy profile in which both players never fight is an SPE.

The same logic as Remark 1 proves the equilibrium uniqueness: no other equilibrium is possible given that (i) not attacking is strictly more beneficial when the other player is following the peaceful strategy (in which she never attacks the opponent) and (ii) mutual attacks do not occur under Assumption 1. □

A.3 Proof of Proposition 1

Proof. I prove the proposition in three simple steps. Before doing so, for simplicity, let us introduce the following notation. Notice that the game is symmetric except for variables related to Z. Then, for a variable/parameter $x \in \{Z_t, z_t, \mu\}$, denote its relevant value to each player by $\tilde{x}_i \equiv \mathbb{1}\{i=1\} \cdot x + \mathbb{1}\{i=2\}(1-x)$. For example, $\tilde{z}_i\theta$ represents the remaining amount of resources for Player 1 (i.e., $z_t\theta$) and Player 2 (i.e., $(1-z_t)\theta$) in a disaster period.

Step 1 (Preliminaries). First, we identify the players' expected payoffs when they follow strategies (S-I) and (S-II). First, consider war payoffs. Suppose that Nature has determined the presence/absence of a disaster and its asymmetric exposure z_t if one occurred. Let $\mathcal{W}_i^{\mathtt{d}}(z_t)$ and $\mathcal{W}_i^{\neg\mathtt{d}}$ denote player i's expected war payoff in a disaster and no-disaster periods, respectively. Then, we have

$$\begin{aligned}
\mathcal{W}_i^{\mathtt{d}}(z_t) &= p_i^{\mathtt{d}}(z_t)\left(\theta(1-c)+\delta\overline{\mathcal{V}}\right) \\
&= \tilde{z}_i\theta\left[\frac{1+\delta(1-\pi)}{1-\delta}-c\right] \\
\mathcal{W}_i^{\neg\mathtt{d}} &= p_i^{\neg\mathtt{d}}\left(2\theta(1-c)+\delta\overline{\mathcal{V}}\right) \\
&= \theta\left[\frac{1-\delta\pi/2}{1-\delta}-c\right].
\end{aligned}$$

Next, consider the expected peace payoffs. Let $\mathcal{V}_i^{\mathtt{S}}$ denote player i's continuation payoff when both players follow strategies (S-I) and (S-II). We can see that $\mathcal{V}_i^{\mathtt{S}}$ consists of four components: cases where (i) a disaster does not take place; (ii) a disaster erupts and neither

player fights; (iii) a disaster takes place and z_t is small so that Player 2 attacks 1; and (iv) a disaster occurs and z_t is large so that Player 1 attacks 2. Thus, we have

$$\begin{aligned}
\mathcal{V}_i^{\mathtt{S}} &= \overbrace{(1-\pi)\left(\theta+\delta\mathcal{V}_i^{\mathtt{S}}\right)}^{(i)\text{ No-disaster period}} \\
&\quad + \underbrace{\pi\left[\begin{array}{l}
\overbrace{\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)}^{(ii)\text{ Pr(Peace)}}\left(\mathbb{E}\left[\tilde{Z}_i\middle|Z_t\in\left[\underline{z}^{\mathtt{S}},\overline{z}^{\mathtt{S}}\right]\right]\theta+\delta\mathcal{V}_i^{\mathtt{S}}\right) \\
+\quad \underbrace{F\left(\underline{z}^{\mathtt{S}}\right)}_{(iii)\text{ Pr(P2 attacks)}}\underbrace{\mathbb{E}\left[\tilde{Z}_i\middle|Z_t<\underline{z}^{\mathtt{S}}\right]\theta\left[(1-c)+\delta\frac{2-\pi}{1-\delta}\right]}_{\mathbb{E}\left[\mathcal{W}_i^{\mathtt{d}}(Z_t)\middle|Z_t<\underline{z}^{\mathtt{S}}\right]} \\
+\ \underbrace{\left(1-F\left(\overline{z}^{\mathtt{S}}\right)\right)}_{(iv)\text{ Pr(P1 attacks)}}\underbrace{\mathbb{E}\left[\tilde{Z}_i\middle|Z_t>\overline{z}^{\mathtt{S}}\right]\theta\left[(1-c)+\delta\frac{2-\pi}{1-\delta}\right]}_{\mathbb{E}\left[\mathcal{W}_i^{\mathtt{d}}(Z_t)\middle|Z_t>\overline{z}^{\mathtt{S}}\right]}
\end{array}\right]}_{\text{Disaster period}},
\end{aligned}$$

where $\Delta_F(b,a)=F(b)-F(a)$ with $a\le b$. Solving this for $\mathcal{V}_i^{\mathtt{S}}$ yields

$$\begin{aligned}
\mathcal{V}_i^{\mathtt{S}} &= \theta\cdot\frac{(1-\pi)+\pi\left[\begin{array}{l}
\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\mathbb{E}\left[\tilde{Z}_i\middle|Z_t\in\left[\underline{z}^{\mathtt{S}},\overline{z}^{\mathtt{S}}\right]\right] \\
+\left(F\left(\underline{z}^{\mathtt{S}}\right)\mathbb{E}\left[\tilde{Z}_i\middle|Z_t<\underline{z}^{\mathtt{S}}\right]+\left(1-F\left(\overline{z}^{\mathtt{S}}\right)\right)\mathbb{E}\left[\tilde{Z}_i\middle|Z_t>\overline{z}^{\mathtt{S}}\right]\right)\left[(1-c)+\delta\frac{2-\pi}{1-\delta}\right]
\end{array}\right]}{1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\right)\right)} \\
&= \theta\cdot\frac{(1-\pi)+\pi\left[\begin{array}{l}
\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\mathbb{E}\left[\tilde{Z}_i\middle|Z_t\in\left[\underline{z}^{\mathtt{S}},\overline{z}^{\mathtt{S}}\right]\right] \\
+F\left(\underline{z}^{\mathtt{S}}\right)\mathbb{E}\left[\tilde{Z}_i\middle|Z_t<\underline{z}^{\mathtt{S}}\right]+\left(1-F\left(\overline{z}^{\mathtt{S}}\right)\right)\mathbb{E}\left[\tilde{Z}_i\middle|Z_t>\overline{z}^{\mathtt{S}}\right] \\
+\left(F\left(\underline{z}^{\mathtt{S}}\right)\mathbb{E}\left[\tilde{Z}_i\middle|Z_t<\underline{z}^{\mathtt{S}}\right]+\left(1-F\left(\overline{z}^{\mathtt{S}}\right)\right)\mathbb{E}\left[\tilde{Z}_i\middle|Z_t>\overline{z}^{\mathtt{S}}\right]\right)\left[\delta\frac{2-\pi}{1-\delta}-c\right]
\end{array}\right]}{1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\right)\right)} \\
&= \theta\cdot\frac{(1-\pi)+\pi\left[\tilde{\mu}_i+\left(F\left(\underline{z}^{\mathtt{S}}\right)\mathbb{E}\left[\tilde{Z}_i\middle|Z_t<\underline{z}^{\mathtt{S}}\right]+\left(1-F\left(\overline{z}^{\mathtt{S}}\right)\right)\mathbb{E}\left[\tilde{Z}_i\middle|Z_t>\overline{z}^{\mathtt{S}}\right]\right)\left[\delta\frac{2-\pi}{1-\delta}-c\right]\right]}{1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\right)\right)},
\end{aligned}$$

where the last equation holds by the law of iterated expectations. By $\mathbb{E}\left[Z_t|Z_t\in[\underline{z},\overline{z}]\right]=$

$\frac{\int_{\underline{z}}^{\overline{z}} z f(z)\mathrm{d}z}{\int_{\underline{z}}^{\overline{z}} f(z)\mathrm{d}z}$, we obtain

$$\mathcal{V}_i^{\mathtt{S}} = \theta \cdot \frac{(1-\pi)+\pi\left(\tilde{\mu}_i+\left(\int_0^{\underline{z}^{\mathtt{S}}}\tilde{z}_i\mathrm{d}F(z)+\int_{\overline{z}^{\mathtt{S}}}^1\tilde{z}_i\mathrm{d}F(z)\right)\left[\delta\frac{2-\pi}{1-\delta}-c\right]\right)}{1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\right)\right)}.$$

When players follow strategies (S-I) and (S-II), player i's expected peace payoff at any information set in a no-disaster period is $\theta+\delta\mathcal{V}_i^{\mathtt{S}}$. On the other hand, i's expected peace payoff in a disaster period when the realized asymmetric exposure is z_t is $\tilde{z}_i\theta+\delta\mathcal{V}_i^{\mathtt{S}}$.

Step 2 (S-I). Consider an arbitrary disaster period. The set of actions (S-I) states that when the realized asymmetric exposure z_t is extreme, the (temporarily) advantaged side attacks the other. Given that the opponent is playing (S-I) and (S-II), if $\mathcal{W}_i^{\mathtt{d}}(z_t)>\tilde{z}_i\theta+\delta\mathcal{V}_i^{\mathtt{S}}$, it is the best response for player i to attack the other in a disaster period t after any history. The above inequality yields

$$\begin{aligned}\tilde{z}_i\theta\left[\frac{1+\delta(1-\pi)}{1-\delta}-c\right] &> \tilde{z}_i\theta+\delta\theta\cdot\frac{(1-\pi)+\pi\left(\tilde{\mu}_i+\left(\int_0^{\underline{z}^{\mathtt{S}}}\tilde{z}_i\mathrm{d}F(z)+\int_{\overline{z}^{\mathtt{S}}}^1\tilde{z}_i\mathrm{d}F(z)\right)\left[\delta\frac{2-\pi}{1-\delta}-c\right]\right)}{1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\right)\right)}\\ \tilde{z}_i &> \delta\cdot\frac{(1-\pi)+\pi\left(\tilde{\mu}_i+\left(\int_0^{\underline{z}^{\mathtt{S}}}\tilde{z}_i\mathrm{d}F(z)+\int_{\overline{z}^{\mathtt{S}}}^1\tilde{z}_i\mathrm{d}F(z)\right)\left[\delta\frac{2-\pi}{1-\delta}-c\right]\right)}{\left(1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\right)\right)\right)\left[\delta\frac{2-\pi}{1-\delta}-c\right]}.\end{aligned}$$

In the above expression, recall that $\tilde{z}_i=z$ and $\tilde{\mu}_i=\mu$ for Player 1. Hence, the threshold $\overline{z}^{\mathtt{S}}$ above which Player 1 attacks 2 is given implicitly by

$$\overline{z}^{\mathtt{S}}=\delta\cdot\frac{(1-\pi)+\pi\left(\mu+\left(\int_0^{\underline{z}^{\mathtt{S}}}z\mathrm{d}F(z)+\int_{\overline{z}^{\mathtt{S}}}^1 z\mathrm{d}F(z)\right)\left[\delta\frac{2-\pi}{1-\delta}-c\right]\right)}{\left(1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\right)\right)\right)\left[\delta\frac{2-\pi}{1-\delta}-c\right]}.$$

On the other hand, by $\tilde{z}_i=1-z$ and $\tilde{\mu}_i=1-\mu$ for Player 2, $\underline{z}^{\mathtt{S}}$ below which Player 2 attacks

1 is implicitly characterized by,

$$\underline{z}^{\mathtt{S}} = 1 - \delta \cdot \frac{(1-\pi) + \pi\left(1 - \mu + \left(1 - \Delta_F\left(\overline{z}^{\mathtt{S}}, \underline{z}^{\mathtt{S}}\right) - \left(\int_0^{\underline{z}^{\mathtt{S}}} z\mathrm{d}F(z) + \int_{\overline{z}^{\mathtt{S}}}^1 z\mathrm{d}F(z)\right)\right)\left[\delta\frac{2-\pi}{1-\delta} - c\right]\right)}{\left(1 - \delta\left(1 - \pi\left(1 - \Delta_F\left(\overline{z}^{\mathtt{S}}, \underline{z}^{\mathtt{S}}\right)\right)\right)\right)\left[\delta\frac{2-\pi}{1-\delta} - c\right]}.$$

Solving the system of the two equations for $\overline{z}^{\mathtt{S}}$ and $\underline{z}^{\mathtt{S}}$ determines the thresholds in realized asymmetric exposure above and below which either one player attacks the other, respectively.

Step 3 (S-II). Finally, consider a no-disaster period. Neither player attacks the opponent when $\mathcal{W}_i^{\neg \mathtt{d}} \leq \theta + \delta \mathcal{V}_i^{\mathtt{S}}$, i.e.,

$$\frac{1}{2}\left[2\theta(1-c) + \delta\frac{\theta(2-\pi)}{1-\delta}\right] \quad \leq \quad \theta + \delta\theta\frac{(1-\pi) + \pi\left(\tilde{\mu}_i + \left(\int_0^{\underline{z}^{\mathtt{S}}} \tilde{z}_i\mathrm{d}F(z) + \int_{\overline{z}^{\mathtt{S}}}^1 \tilde{z}_i\mathrm{d}F(z)\right)\left[\delta\frac{2-\pi}{1-\delta} - c\right]\right)}{1 - \delta\left(1 - \pi\left(1 - \Delta_F\left(\overline{z}^{\mathtt{S}}, \underline{z}^{\mathtt{S}}\right)\right)\right)}.$$

Rearranging this yields

$$\begin{aligned}\tilde{\mu}_i \quad \geq \quad & 1 - \frac{1}{\pi} + \delta\frac{2-\pi}{1-\delta}\left[\frac{1 - \delta\left(1 - \pi\left(1 - \Delta_F\left(\overline{z}^{\mathtt{S}}, \underline{z}^{\mathtt{S}}\right)\right)\right)}{2\delta\pi} - \left(\int_0^{\underline{z}^{\mathtt{S}}} \tilde{z}_i\mathrm{d}F(z) + \int_{\overline{z}^{\mathtt{S}}}^1 \tilde{z}_i\mathrm{d}F(z)\right)\right] \\ & -c\left[\frac{1 - \delta\left(1 - \pi\left(1 - \Delta_F\left(\overline{z}^{\mathtt{S}}, \underline{z}^{\mathtt{S}}\right)\right)\right)}{\delta\pi} - \left(\int_0^{\underline{z}^{\mathtt{S}}} \tilde{z}_i\mathrm{d}F(z) + \int_{\overline{z}^{\mathtt{S}}}^1 \tilde{z}_i\mathrm{d}F(z)\right)\right].\end{aligned}$$

By $\frac{1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathtt{S}}, \underline{z}^{\mathtt{S}}\right)\right)\right)}{2\delta\pi} = \frac{1-F\left(\overline{z}^{\mathtt{S}}\right)+F\left(\underline{z}^{\mathtt{S}}\right)}{2} + \frac{1-\delta}{2\delta\pi}$, factoring out the second term leads to

$$\begin{aligned}\tilde{\mu}_i \quad \geq \quad & \frac{1}{2} + \delta\frac{2-\pi}{1-\delta}\left(\int_0^{\underline{z}^{\mathtt{S}}} \frac{1}{2} - \tilde{z}_i\mathrm{d}F(z) + \int_{\overline{z}^{\mathtt{S}}}^1 \frac{1}{2} - \tilde{z}_i\mathrm{d}F(z)\right) \\ & -c\left(\int_0^{\underline{z}^{\mathtt{S}}} 1 - \tilde{z}_i\mathrm{d}F(z) + \int_{\overline{z}^{\mathtt{S}}}^1 1 - \tilde{z}_i\mathrm{d}F(z) + \frac{1-\delta}{\delta\pi}\right),\end{aligned}$$

where $\tilde{\mu}_i$ for which the above expression holds with equality is the threshold value. Note that we have a system of three equations with three unknowns (i.e., $\underline{z}^{\mathtt{S}}(\mu)$, $\overline{z}^{\mathtt{S}}(\mu)$, and μ). By solving the system of equations, by $\tilde{\mu}_i = \mathbb{1}\{i=1\} \cdot \mu + \mathbb{1}\{i=2\}(1-\mu)$ and $\tilde{z}_i =$

$\mathbb{1}\{i=1\}\cdot z+\mathbb{1}\{i=2\}(1-z)$, we obtain the desired result. □

A.4 Proof of Proposition 2

Proof. We can prove this proposition in a similar manner to Proposition 1.

Step 1 (Preliminaries). From the proof of Proposition 1, we know players' expected war payoffs, i.e., $\mathcal{W}_i^{\mathtt{d}}(z_t)$ and $\mathcal{W}_i^{\neg\mathtt{d}}$. Let $\mathcal{V}_i^{\mathtt{A}}$ represent player i's continuation payoff when both players are following strategies (A-I) and (A-II):

$$\mathcal{V}_i^{\mathtt{A}} = \overbrace{(1-\pi)\,\mathcal{W}_i^{\neg\mathtt{d}}}^{\text{No-disaster period}} + \underbrace{\pi\left[\begin{array}{l} \overbrace{\Delta_F\left(\overline{z}^{\mathtt{A}},\underline{z}^{\mathtt{A}}\right)}^{\Pr(\text{Peace})}\left(\mathbb{E}\left[\tilde{Z}_i\middle|Z_t\in\left[\underline{z}^{\mathtt{A}},\overline{z}^{\mathtt{A}}\right]\right]\theta+\delta\mathcal{V}_i^{\mathtt{A}}\right) \\ \quad+\underbrace{F\left(\underline{z}^{\mathtt{A}}\right)}_{\Pr(\text{P2 attacks})}\underbrace{\mathbb{E}\left[\tilde{Z}_i\middle|Z_t<\underline{z}^{\mathtt{A}}\right]\theta\left[(1-c)+\delta\frac{2-\pi}{1-\delta}\right]}_{\mathbb{E}\left[\mathcal{W}_i^{\mathtt{d}}(Z_t)\middle|Z_t<\underline{z}^{\mathtt{A}}\right]} \\ \quad+\underbrace{\left(1-F\left(\overline{z}^{\mathtt{A}}\right)\right)}_{\Pr(\text{P1 attacks})}\underbrace{\mathbb{E}\left[\tilde{Z}_i\middle|Z_t>\overline{z}^{\mathtt{A}}\right]\theta\left[(1-c)+\delta\frac{2-\pi}{1-\delta}\right]}_{\mathbb{E}\left[\mathcal{W}_i^{\mathtt{d}}(Z_t)\middle|Z_t>\overline{z}^{\mathtt{A}}\right]} \end{array}\right]}_{\text{Disaster period}}.$$

Solving this for $\mathcal{V}_i^{\mathtt{A}}$, we obtain

$$\mathcal{V}_i^{\mathtt{A}}=\theta\cdot\frac{(1-\pi)\left[\dfrac{1-\delta\pi/2}{1-\delta}-c\right]+\pi\left(\tilde{\mu}_i+\left(\displaystyle\int_0^{\underline{z}^{\mathtt{A}}}\tilde{z}_i\mathrm{d}F(z)+\int_{\overline{z}^{\mathtt{A}}}^1\tilde{z}_i\mathrm{d}F(z)\right)\left[\delta\dfrac{2-\pi}{1-\delta}-c\right]\right)}{1-\delta\pi\Delta_F\left(\overline{z}^{\mathtt{A}},\underline{z}^{\mathtt{A}}\right)}.$$

Step 2 (A-I). Consider an arbitrary disaster period t and suppose that the opponent is playing strategies (A-I) and (A-II). Then, it is optimal for player i to attack the other when

$\mathcal{W}_i^{\mathtt{d}}(z_t) > \tilde{z}_i\theta + \delta\mathcal{V}_i^{\mathtt{A}}$. That is,

$$\begin{aligned}
\tilde{z}_i\theta\left[\frac{1+\delta(1-\pi)}{1-\delta}-c\right] &> \tilde{z}_i\theta + \delta\theta\cdot\frac{\left[\begin{array}{l}(1-\pi)\left[\dfrac{1-\delta\pi/2}{1-\delta}-c\right]+\\ \pi\left(\tilde{\mu}_i+\left(\displaystyle\int_0^{\underline{z}^{\mathtt{A}}}\tilde{z}_i\mathrm{d}F(z)+\int_{\overline{z}^{\mathtt{A}}}^1\tilde{z}_i\mathrm{d}F(z)\right)\left[\delta\dfrac{2-\pi}{1-\delta}-c\right]\right)\end{array}\right]}{1-\delta\pi\Delta_F\left(\overline{z}^{\mathtt{A}},\underline{z}^{\mathtt{A}}\right)}\\
\tilde{z}_i &> \delta\cdot\frac{\left[\begin{array}{l}(1-\pi)\left[\dfrac{1-\delta\pi/2}{1-\delta}-c\right]+\\ \pi\left(\tilde{\mu}_i+\left(\displaystyle\int_0^{\underline{z}^{\mathtt{A}}}\tilde{z}_i\mathrm{d}F(z)+\int_{\overline{z}^{\mathtt{A}}}^1\tilde{z}_i\mathrm{d}F(z)\right)\left[\delta\dfrac{2-\pi}{1-\delta}-c\right]\right)\end{array}\right]}{\left(1-\delta\pi\Delta_F\left(\overline{z}^{\mathtt{A}},\underline{z}^{\mathtt{A}}\right)\right)\left[\delta\dfrac{2-\pi}{1-\delta}-c\right]}.
\end{aligned}$$

By $\tilde{z}_i = z_t$ for player 1 and $\tilde{z}_i = 1 - z_t$ for player 2, substituting them in the above inequality yields the thresholds $\underline{z}^{\mathtt{A}}$ and $\overline{z}^{\mathtt{A}}$.

Step 3 (A-II). Similarly, consider an arbitrary no-disaster period. When the other player is playing strategies (A-I) and (A-II), it is optimal to attack the other when $\mathcal{W}_i^{\neg\mathtt{d}} > \theta + \delta\mathcal{V}_i^{\mathtt{A}}$. Namely,

$$\begin{aligned}
\frac{1}{2}\left[2\theta(1-c)+\delta\frac{\theta(2-\pi)}{1-\delta}\right] &> \theta + \delta\theta\cdot\frac{\left[\begin{array}{l}(1-\pi)\left[\dfrac{1-\delta\pi/2}{1-\delta}-c\right]+\\ \pi\left(\tilde{\mu}_i+\left(\displaystyle\int_0^{\underline{z}^{\mathtt{A}}}\tilde{z}_i\mathrm{d}F(z)+\int_{\overline{z}^{\mathtt{A}}}^1\tilde{z}_i\mathrm{d}F(z)\right)\left[\delta\dfrac{2-\pi}{1-\delta}-c\right]\right)\end{array}\right]}{1-\delta\pi\Delta_F\left(\overline{z}^{\mathtt{A}},\underline{z}^{\mathtt{A}}\right)}\\
\tilde{\mu}_i &< \frac{1}{2}+\delta\frac{2-\pi}{1-\delta}\left(\int_0^{\underline{z}^{\mathtt{A}}}\frac{1}{2}-\tilde{z}_i\mathrm{d}F(z)+\int_{\overline{z}^{\mathtt{A}}}^1\frac{1}{2}-\tilde{z}_i\mathrm{d}F(z)\right)\\
&\quad -c\left(\int_0^{\underline{z}^{\mathtt{A}}}1-\tilde{z}_i\mathrm{d}F(z)+\int_{\overline{z}^{\mathtt{A}}}^1 1-\tilde{z}_i\mathrm{d}F(z)+\frac{1-\delta}{\delta\pi}\right).
\end{aligned}$$

As in the case of Proposition 1, we obtain the desired results by $\tilde{\mu}_i = \mu$ and $\tilde{z}_i = z$ for Player 1 and $\tilde{\mu}_i = 1 - \mu$ and $\tilde{z}_i = 1 - z$ for Player 2. □

A.5 Proof of Proposition 3

To prove Proposition 3, consider the following lemmas.

Lemma 1. *Under Assumption 1, there is no mixed-strategy equilibrium.*

Proof. This directly stems from the fact that players are indifferent between pure strategies in the support of any mixed (equilibrium) strategy. Namely, if a player were to mix attacking and not attacking at some t, she must be indifferent between the two actions. Assumption 1 assigns "not attack" in this case, which rules out mixed-strategy equilibria. □

Lemma 2. *The set of values of z_t under which a player fights when $D_t = 1$ constitutes a one-sided interval.*

Proof. We know that $\mathcal{W}_i^{\mathtt{d}}(z_t)$ is monotonic in z_t: it is strictly increasing and decreasing for player 1 and player 2, respectively. Let $\mathcal{V}_i$ represent a continuation payoff generated by arbitrary equilibrium strategies. Also recall we have denoted $\tilde{z}_i = \mathbb{1}\{i=1\}\cdot z_t + \mathbb{1}\{i=2\}(1-z_t)$. Then, player i attacks the other in this equilibrium at period t such that $D_t = 1$ when

$$\begin{aligned} \tilde{z}_i\theta\left[(1-c)+\delta\frac{2-\pi}{1-\delta}\right] &> \tilde{z}_i\theta + \delta\mathcal{V}_i \\ \tilde{z}_i &> \frac{\delta\mathcal{V}_i}{\theta\left[\delta\frac{2-\pi}{1-\delta}-c\right]}. \end{aligned}$$

Note that the continuation payoff $\mathcal{V}_i$ is not a function of the current realized damage $\tilde{z}_i$. Hence, whenever $\frac{\delta\mathcal{V}_i}{\theta\left[\delta\frac{2-\pi}{1-\delta}-c\right]} \in (0,1)$, we can conclude that the set of values of $\tilde{z}_i$ under which i fights is $\left(\frac{\delta\mathcal{V}_i}{\theta\left[\delta\frac{2-\pi}{1-\delta}-c\right]}, 1\right]$, which is a one-sided interval. □

Thus, we have established that, in equilibrium, players follow cutoff strategies when $D_t = 1$. Let $\overline{z}$ be Player 1's cutoff in Z_t above which she attacks Player 2 in a disaster period, and analogously, $\underline{z}$ be Player 2's cutoff.

Lemma 3. *The cutoff $\overline{z}$ is weakly decreasing in $\underline{z}$, and $\underline{z}$ is weakly decreasing in $\overline{z}$.*

Proof. Focus on Player 1's decision-making. Suppose that $\underline{z}$ strictly increases from some $\underline{z}'$ to $\underline{z}''$, i.e., Player 2 is more willing to fight after a disaster. Such an increase must weakly reduce $\mathcal{V}_1$. To see this, assume $Z_t = \hat{z} \in (\underline{z}', \underline{z}'')$. There are two possible cases of how this increase in $\underline{z}$ affects Player 1's continuation payoff. First, when Player 1 would have preferred peace and would not have otherwise fought after observing $\hat{z}$ under $\underline{z}'$, then Player 2's attack as a result of the higher $\underline{z}''$ must lower Player 1's continuation payoff. Second, when Player 1 would fight after seeing $\hat{z}$ in any case, then Player 2's higher willingness to fight does not affect Player 1's payoff. Thus, $\mathcal{V}_1$ must be weakly decreasing in $\underline{z}$.

Now recall that the cutoff $\overline{z}$ is defined as the value of Z_t such that $\overline{z}\theta + \delta\mathcal{V}_1 = \overline{z}\left[\theta(1-c) + \delta\overline{\mathcal{V}}\right]$ holds, which yields $\overline{z} = \frac{\delta\mathcal{V}_1}{\theta\left[\delta\frac{2-\pi}{1-\delta} - c\right]}$. We have shown that $\mathcal{V}_1$ is weakly decreasing in $\underline{z}$, and this equality shows that a change in $\underline{z}$ affects the optimal $\overline{z}$ only through $\mathcal{V}_1$. Consequently, $\overline{z}$ is weakly decreasing in $\underline{z}$. An analogous logic shows that $\underline{z}$ is decreasing in $\overline{z}$. □

Lemma 4. *Fix parameter values and F. Then, we have*

$$\underline{\mu}^{\mathcal{S}} < \underline{\mu}^{\mathcal{A}} \quad \textit{and} \quad \overline{\mu}^{\mathcal{A}} < \overline{\mu}^{\mathcal{S}}.$$

Proof. Focus on $\underline{\mu}^{\mathtt{S}} < \underline{\mu}^{\mathtt{A}}$, which are Player 1's thresholds. We first show that the inequality holds with a weak inequality, then show that the intervals must overlap with a nonsingleton intersection. When $\underline{\mu}^{\mathtt{S}} = \underline{\mu}^{\mathtt{A}}$, we denote the common threshold by $\underline{\mu}^*$.

Step 1 ($\underline{\mu}^{\mathtt{S}} \leq \underline{\mu}^{\mathtt{A}}$). Suppose to the contrary that $\underline{\mu}^{\mathtt{S}} > \underline{\mu}^{\mathtt{A}}$, which implies there is a gap between these thresholds. Consider $\underline{\mu}^{\mathtt{S}}$. Denote Player 1's continuation payoff at $\underline{\mu}^{\mathtt{S}}$ in the case of ($\mathtt{S}$) by $\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^{\mathtt{S}}\right)$. By the definition of $\underline{\mu}^{\mathtt{S}}$, we obtain $\mathcal{W}_1^{\neg\mathtt{d}} = \theta + \delta\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^{\mathtt{S}}\right)$. Hence, for arbitrarily small $\varepsilon > 0$, it follows that $\mathcal{W}_1^{\neg\mathtt{d}} > \theta + \delta\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^{\mathtt{S}} - \varepsilon\right)$. Hence, Player 1 fights at any t such that $D_t = 0$ when $\mu = \underline{\mu}^{\mathtt{S}} - \varepsilon$. Player 2 does not fight either because she prefers peace or is indifferent (by Assumption 1). By Lemma 2, we already know that players must follow cutoff strategies when $D_t = 1$. In sum, (i) Player 1 fights whenever $D_t = 0$, (ii) Player 2 does not whenever $D_t = 0$, and (iii) they employ cutoff strategies when $D_t = 1$. This is a

strategy profile ($\mathtt{A}_1$). Finally, observe that this strategy profile at $\underline{\mu}^{\mathtt{S}} - \varepsilon$ indeed constitutes a stationary equilibrium. That is, (i) Player 1 always fights and Player 2 never does in any t such that $D_t = 0$, and (ii) there exists a pair of cutoff strategies that are a best response to each other (Proposition 5 in Appendix B). Thus, there cannot be a gap between $M^{\mathtt{S}}$ and $M^{\mathtt{A}_1}$.

Step 2 ($\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right) \geq \lim_{\mu \nearrow \underline{\mu}^*} \overline{z}^{\mathtt{A}_1}(\mu)$). Suppose $\underline{\mu}^{\mathtt{S}} = \underline{\mu}^{\mathtt{A}} = \underline{\mu}^*$, which implies $M^{\mathtt{S}} \cap M^{\mathtt{A}_1} = \emptyset$. By the definition of $\underline{\mu}^*$, we have $\mathcal{W}_1^{\neg \mathtt{d}} = \theta + \delta \mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right)$. Because we have assumed F is continuous and by the expressions in Proposition 2, we know that Player 1's continuation payoff at μ in the case of ($\mathtt{A}_1$), $\mathcal{V}_1^{\mathtt{A}_1}(\mu)$, is continuous in μ. Thus, $\lim_{\mu \nearrow \underline{\mu}^*} \mathcal{V}_1^{\mathtt{A}_1}(\mu)$ exists. Similarly, $\lim_{\mu \nearrow \underline{\mu}^*} \overline{z}^{\mathtt{A}_1}(\mu)$ also exists. Because $\mathcal{W}_1^{\neg \mathtt{d}} > \theta + \delta \mathcal{V}_1^{\mathtt{A}_1}\left(\mu < \underline{\mu}^*\right)$, we obtain $\mathcal{W}_1^{\neg \mathtt{d}} \geq \theta + \delta \times \lim_{\mu \nearrow \underline{\mu}^*} \mathcal{V}_1^{\mathtt{A}_1}(\mu)$. In sum, we have established $\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right) \geq \lim_{\mu \nearrow \underline{\mu}^*} \mathcal{V}_1^{\mathtt{A}_1}(\mu)$. By Lemma 2, we know that Player 1's cutpoint above which she fights when $D_t = 1$ is increasing in her continuation value. Thus, we obtain $\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right) \geq \lim_{\mu \nearrow \underline{\mu}^*} \overline{z}^{\mathtt{A}_1}(\mu)$.

Step 3 ($\underline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right) < \lim_{\mu \nearrow \underline{\mu}^*} \underline{z}^{\mathtt{A}_1}(\mu)$). By Lemma 3, we know that (i) a player's willingness to fight when $D_t = 1$ (i.e., $1 - \overline{z}$ for Player 1 and $\underline{z}$ for Player 2) is strictly decreasing in her continuation value and that (ii) the continuation value is weakly decreasing in the opponent's willingness to fight when $D_t = 1$. Thus, by observation (ii) above and by $\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right) \geq \lim_{\mu \nearrow \underline{\mu}^*} \overline{z}^{\mathtt{A}_1}(\mu)$ shown in Step 2, Player 2's continuation value has $\mathcal{V}_2^{\mathtt{S}}\left(\underline{\mu}^*\right) \geq \lim_{\mu \nearrow \underline{\mu}^*} \mathcal{V}_2^{\mathtt{A}_1}(\mu)$. Next, consider t such that $D_t = 0$. Recall that $\underline{\mu}^*$ is the point at which $\mathcal{W}_1^{\neg \mathtt{d}} = \theta + \delta \mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right)$ holds for Player 1. Therefore, for Player 2, it follows that $\mathcal{W}_2^{\neg \mathtt{d}} < \theta + \delta \mathcal{V}_2^{\mathtt{S}}\left(\underline{\mu}^*\right)$. Thus, if Player 1 fights when $D_t = 0$, it strictly decreases Player 2's continuation payoff. In sum, we obtain $\mathcal{V}_2^{\mathtt{S}}\left(\underline{\mu}^*\right) > \lim_{\mu \nearrow \underline{\mu}^*} \mathcal{V}_2^{\mathtt{A}_1}(\mu)$. By observation (i) above, we have established $\underline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right) < \lim_{\mu \nearrow \underline{\mu}^*} \underline{z}^{\mathtt{A}_1}(\mu)$.

Step 4 ($\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right) > \lim_{\mu \nearrow \underline{\mu}^*} \mathcal{V}_1^{\mathtt{A}_1}(\mu)$). For simplicity, for $\mathtt{E} \in \{\mathtt{S}, \mathtt{A}_1\}$, denote $\Delta^{\mathtt{E}} \equiv \Delta_F\left(\overline{z}^{\mathtt{E}}\left(\underline{\mu}^*\right), \underline{z}^{\mathtt{E}}\left(\underline{\mu}^*\right)\right)$ and $\Sigma^{\mathtt{E}} \equiv \int_0^{\underline{z}^{\mathtt{E}}\left(\underline{\mu}^*\right)} z \mathrm{d}F(z) + \int_{\overline{z}^{\mathtt{E}}\left(\underline{\mu}^*\right)}^1 z \mathrm{d}F(z)$, where the thresholds are

left-sided limits as $\mu \nearrow \underline{\mu}^*$ in the case of $(\mathtt{A}_1)$. Then, observe

$$\begin{aligned}
\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right) &= (1-\pi)\left(\theta + \delta \mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right)\right) + \pi\left(\theta\underline{\mu}^* + \theta\Sigma^{\mathtt{S}}\left[\delta\frac{2-\pi}{1-\delta} - c\right] + \delta\Delta^{\mathtt{S}}\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right)\right) \\
\lim_{\mu\nearrow\underline{\mu}^*} \mathcal{V}_1^{\mathtt{A}_1}(\mu) &= (1-\pi)\mathcal{W}_1^{\neg\mathtt{d}} + \pi\left(\theta\underline{\mu}^* + \theta\Sigma^{\mathtt{A}_1}\left[\delta\frac{2-\pi}{1-\delta} - c\right] + \delta\Delta^{\mathtt{A}_1}\lim_{\mu\nearrow\underline{\mu}^*}\mathcal{V}_1^{\mathtt{A}_1}(\mu)\right),
\end{aligned}$$

where we have $\theta\underline{\mu}^*$ in the parentheses, as in Propositions 1 and 2, by the law of iterated expectations. We evaluate the difference between the two continuation values, $\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right) - \lim_{\mu\nearrow\underline{\mu}^*}\mathcal{V}_1^{\mathtt{A}_1}(\mu)$. First, by the definition of $\underline{\mu}^*$ where Player 1 becomes indifferent between fighting and not fighting when $D_t = 0$, note that $\left(\theta + \delta\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right)\right) - \mathcal{W}_1^{\neg\mathtt{d}} = 0$. Thus, we have

$$\begin{aligned}
\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right) - \lim_{\mu\nearrow\underline{\mu}^*}\mathcal{V}_1^{\mathtt{A}_1}(\mu) &= \pi\left(\theta\left[\delta\frac{2-\pi}{1-\delta} - c\right]\left(\Sigma^{\mathtt{S}} - \Sigma^{\mathtt{A}_1}\right) + \delta\left(\Delta^{\mathtt{S}}\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right) - \Delta^{\mathtt{A}_1}\lim_{\mu\nearrow\underline{\mu}^*}\mathcal{V}_1^{\mathtt{A}_1}(\mu)\right)\right) \\
&= \pi\left(\begin{array}{l}\theta\left[\delta\frac{2-\pi}{1-\delta} - c\right]\left(\Sigma^{\mathtt{S}} - \Sigma^{\mathtt{A}_1}\right) \\ +\delta\left(\Delta^{\mathtt{S}}\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right)\underbrace{-\Delta^{\mathtt{A}_1}\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right) + \Delta^{\mathtt{A}_1}\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right)}_{+0} - \Delta^{\mathtt{A}_1}\lim_{\mu\nearrow\underline{\mu}^*}\mathcal{V}_1^{\mathtt{A}_1}(\mu)\right)\end{array}\right).
\end{aligned}$$

Factoring $\delta\pi\Delta^{\mathtt{A}_1}\left(\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right) - \lim_{\mu\nearrow\underline{\mu}^*}\mathcal{V}_1^{\mathtt{A}_1}(\mu)\right)$ out, this yields

$$\begin{aligned}
\left(\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right) - \lim_{\mu\nearrow\underline{\mu}^*}\mathcal{V}_1^{\mathtt{A}_1}(\mu)\right)\left(1 - \delta\pi\Delta^{\mathtt{A}_1}\right) &= \pi\left(\theta\left[\delta\frac{2-\pi}{1-\delta} - c\right]\left(\Sigma^{\mathtt{S}} - \Sigma^{\mathtt{A}_1}\right) + \delta\left(\Delta^{\mathtt{S}} - \Delta^{\mathtt{A}_1}\right)\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right)\right) \\
&= \pi\theta\left[\delta\frac{2-\pi}{1-\delta} - c\right]\left(\left(\Sigma^{\mathtt{S}} - \Sigma^{\mathtt{A}_1}\right) + \overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)\left(\Delta^{\mathtt{S}} - \Delta^{\mathtt{A}_1}\right)\right)
\end{aligned}$$

where second equality holds by $\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right) = \frac{\delta\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right)}{\theta\left[\delta\frac{2-\pi}{1-\delta} - c\right]}$. Because $1 - \delta\pi\Delta^{\mathtt{A}_1} > 0$, we have $\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right) >$

$\lim_{\mu \nearrow \underline{\mu}^*} \mathcal{V}_1^{\mathtt{A}_1}(\mu)$ if the right-hand side is positive:

$$
\begin{aligned}
& \pi\theta\left[\delta\frac{2-\pi}{1-\delta}-c\right]\left(\left(\Sigma^{\mathtt{S}}-\Sigma^{\mathtt{A}_1}\right)+\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)\left(\Delta^{\mathtt{S}}-\Delta^{\mathtt{A}_1}\right)\right) \\
= \quad & \pi\theta\left[\delta\frac{2-\pi}{1-\delta}-c\right]\left(\begin{array}{l}
\displaystyle\int_0^{\underline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)} z\mathrm{d}F(z)+\int_{\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)}^1 z\mathrm{d}F(z) \\
\displaystyle-\int_0^{\lim_{\mu\nearrow\underline{\mu}^*}\underline{z}^{\mathtt{A}_1}(\mu)} z\mathrm{d}F(z)-\int_{\lim_{\mu\nearrow\underline{\mu}^*}\overline{z}^{\mathtt{A}_1}(\mu)}^1 z\mathrm{d}F(z) \\
\displaystyle+\int_0^{\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)}\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)\mathrm{d}F(z)-\int_0^{\underline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)}\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)\mathrm{d}F(z) \\
\displaystyle-\int_0^{\lim_{\mu\nearrow\underline{\mu}^*}\overline{z}^{\mathtt{A}_1}(\mu)}\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)\mathrm{d}F(z)+\int_0^{\lim_{\mu\nearrow\underline{\mu}^*}\underline{z}^{\mathtt{A}_1}(\mu)}\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)\mathrm{d}F(z)
\end{array}\right).
\end{aligned}
$$

By simplifying this, we obtain

$$
\pi\theta\left[\delta\frac{2-\pi}{1-\delta}-c\right]\left(\underbrace{\int_{\underline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)}^{\lim_{\mu\nearrow\underline{\mu}^*}\underline{z}^{\mathtt{A}_1}(\mu)}\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)-z\mathrm{d}F(z)}_{>0}+\underbrace{\int_{\lim_{\mu\nearrow\underline{\mu}^*}\overline{z}^{\mathtt{A}_1}(\mu)}^{\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)}\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)-z\mathrm{d}F(z)}_{\geq 0}\right)>0.
$$

The inequality holds because, first, the first integral in the parentheses is strictly positive, given (i) $\underline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)<\lim_{\mu\nearrow\underline{\mu}^*}\underline{z}^{\mathtt{A}_1}(\mu)$ by Step 3 and (ii) $\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)-z>0$ in this interval. Second, the second integral is nonnegative, given (i) $\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)\geq\lim_{\mu\nearrow\underline{\mu}^*}\overline{z}^{\mathtt{A}_1}(\mu)$ by Step 2 and (ii) $\overline{z}^{\mathtt{S}}\left(\underline{\mu}^*\right)-z\geq 0$ in this interval.

Step 5 ($\underline{\mu}^{\mathtt{S}}<\underline{\mu}^{\mathtt{A}}$). Suppose to the contrary that $\underline{\mu}^{\mathtt{S}}=\underline{\mu}^{\mathtt{A}}=\mu^*$. Because $\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right)>\lim_{\mu\nearrow\underline{\mu}^*}\mathcal{V}_1^{\mathtt{A}_1}(\mu)$ holds, by the definition of $\underline{\mu}^*$, we have $\mathcal{W}_1^{\neg\mathtt{d}}=\theta+\delta\mathcal{V}_1^{\mathtt{S}}\left(\underline{\mu}^*\right)>\theta+\delta\lim_{\mu\nearrow\underline{\mu}^*}\mathcal{V}_1^{\mathtt{A}_1}(\mu)$. Hence, by continuity of $\mathcal{V}_1^{\mathtt{A}_1}(\mu)$ in μ, there exists $\varepsilon>0$ such that, for all $\mu\in\left[\underline{\mu}^*,\underline{\mu}^*+\varepsilon\right)$, $\mathcal{W}_1^{\neg\mathtt{d}}>\theta+\delta\mathcal{V}_1^{\mathtt{A}_1}(\mu)$ holds. This implies that, for μ in the interval, Player 1 strictly prefers fighting when $D_t=0$ under the strategy profile $(\mathtt{A}_1)$. In such a case, observe that Player 2 does not fight when $D_t=0$ either because she strictly prefers peace or because she is indifferent (Assumption 1). By Lemma 2, recall that players have cutoff strategies when $D_t=1$. Thus, for all $\mu\in\left[\underline{\mu}^*,\underline{\mu}^*+\varepsilon\right)$, a strategy profile categorized as $(\mathtt{A}_1)$ still constitutes

an equilibrium. This contradicts the assumption that $\underline{\mu}^* = \underline{\mu}^{\mathtt{S}} = \underline{\mu}^{\mathtt{A}}$. Analogous discussions apply to $\overline{\mu}^{\mathtt{A}} < \overline{\mu}^{\mathtt{S}}$. □

Lemma 5. *Fix all parameters other than μ. When $\overline{\mu}^s$ and $\underline{\mu}^s$ satisfy*

$$c < \min \left\{ \begin{array}{l} \dfrac{\frac{1}{2} - \delta \frac{2-\pi}{1-\delta} \left(\int_0^{\underline{z}^s(\overline{\mu}^s)} \frac{1}{2} - z \mathrm{d}F_{\overline{\mu}^s}(z) + \int_{\overline{z}^s(\overline{\mu}^s)}^1 \frac{1}{2} - z \mathrm{d}F_{\overline{\mu}^s}(z) \right)}{\int_0^{\underline{z}^s(\overline{\mu}^s)} z \mathrm{d}F_{\overline{\mu}^s}(z) + \int_{\overline{z}^s(\overline{\mu}^s)}^1 z \mathrm{d}F_{\overline{\mu}^s}(z) + \dfrac{1-\delta}{\delta\pi}}, \\ \dfrac{\frac{1}{2} + \delta \frac{2-\pi}{1-\delta} \left(\int_0^{\underline{z}^s(\underline{\mu}^s)} \frac{1}{2} - z \mathrm{d}F_{\underline{\mu}^s}(z) + \int_{\overline{z}^s(\underline{\mu}^s)}^1 \frac{1}{2} - z \mathrm{d}F_{\underline{\mu}^s}(z) \right)}{\int_0^{\underline{z}^s(\underline{\mu}^s)} 1 - z \mathrm{d}F_{\underline{\mu}^s}(z) + \int_{\overline{z}^s(\underline{\mu}^s)}^1 1 - z \mathrm{d}F_{\underline{\mu}^s}(z) + \dfrac{1-\delta}{\delta\pi}} \end{array} \right\},$$

we have $\underline{\mu}^s > 0$ and $\overline{\mu}^s < 1$.

Proof. Using expressions in Proposition 1, rearranging inequalities $\underline{\mu}^{\mathtt{S}} > 0$ and $\overline{\mu}^{\mathtt{S}} < 1$ immediately leads to the condition in the lemma. □

Now we are ready to prove Proposition 3.

Proof of Proposition 3. Assumption 1 and Lemma 1 guarantee that, in stationary strategies, each player either fights for sure or never fights in no-disaster periods. This observation and Lemma 2 imply that there are only three combinations of equilibrium strategies in our equilibrium concept (stationary MPE in pure strategies): (i) Neither attacks when $D_t = 0$; (ii) Player 1 attacks when $D_t = 0$; and (iii) Player 2 attacks when $D_t = 0$, all with one-sided intervals of z_t under which players fight when $D_t = 1$. They correspond to strategy profiles $(\mathtt{S})$ $(\mathtt{A}_1)$, and $(\mathtt{A}_2)$, respectively. Lemma 4 shows that there are regions of equilibrium multiplicity, $\left[\underline{\mu}^{\mathtt{S}}, \underline{\mu}^{\mathtt{A}}\right)$ and $(\overline{\mu}^{\mathtt{A}}, \overline{\mu}^{\mathtt{S}}]$. Finally, Lemma 5 implicitly specifies conditions under which the partition exists in $[0, 1]$. □

A.6 Proof of Corollary 1

Proof. This results directly follows from the equilibrium strategies specified in Propositions 1 and 2. By Proposition 1, we have

$$\begin{aligned}
\Pr\left(\text{Conflict at } t \text{ such that } D_t = 1 \text{ in case } (\mathtt{S})\right) &= \Pr\left(Y_t^{\mathtt{game}} = 1 \middle| D_t = 1, \mu^{\mathtt{S}}\right) \\
&\in (0,1) \quad \text{and} \\
\Pr\left(\text{Conflict at } t \text{ such that } D_t = 0 \text{ in case } (\mathtt{S})\right) &= \Pr\left(Y_t^{\mathtt{game}} = 1 \middle| D_t = 0, \mu^{\mathtt{S}}\right) \\
&= 0.
\end{aligned}$$

Similarly, Proposition 2 yields

$$\begin{aligned}
\Pr\left(\text{Conflict at } t \text{ such that } D_t = 1 \text{ in cases } (\mathtt{A}_1) \text{ or } (\mathtt{A}_2)\right) &= \Pr\left(Y_t^{\mathtt{game}} = 1 \middle| D_t = 1, \mu^{\mathtt{A}}\right) \\
&\in (0,1) \quad \text{and} \\
\Pr\left(\text{Conflict at } t \text{ such that } D_t = 0 \text{ in cases } (\mathtt{A}_1) \text{ or } (\mathtt{A}_2)\right) &= \Pr\left(Y_t^{\mathtt{game}} = 1 \middle| D_t = 0, \mu^{\mathtt{A}}\right) \\
&= 1.
\end{aligned}$$

which completes the proof. □

A.7 Proof of Corollary 2

Proof. By Corollary 1 and the identity $Y_{dt}^{\mathtt{data}} = D_{dt}Y_{dt}^{\mathtt{PO}}(1) + (1 - D_{dt})Y_{dt}^{\mathtt{PO}}(0)$, direct substitutions yield part (i) of the statement:

$$\begin{aligned}
& \Pr\left(Y_t^{\mathtt{game}} = 1 \middle| D_t = 1, \mu^{\mathtt{S}}\right) > \Pr\left(Y_t^{\mathtt{game}} = 1 \middle| D_t = 0, \mu^{\mathtt{S}}\right) \\
\overset{\text{Assumption 2}}{\Longrightarrow} \quad & \Pr\left(Y_{dt}^{\mathtt{data}} = 1 \middle| D_{dt} = 1, \mu^{\mathtt{S}}\right) > \Pr\left(Y_{dt}^{\mathtt{data}} = 1 \middle| D_{dt} = 0, \mu^{\mathtt{S}}\right) \\
\Longrightarrow \quad & \Pr\left(Y_{dt}^{\mathtt{PO}}(1) = 1 \middle| \mu^{\mathtt{S}}\right) > \Pr\left(Y_{dt}^{\mathtt{PO}}(0) = 1 \middle| \mu^{\mathtt{S}}\right).
\end{aligned}$$

Part (ii) is derived in the same manner. □

A.8 Proof of Corollary 3

Proof. From the discussion in the main text, we know

$$\tau \in \left[\tau\left(\mu^{\mathtt{S}}\right) q^{\mathtt{S}} + \tau\left(\mu^{\mathtt{A}}\right)\left(1-q^{\mathtt{S}}\right), \tau\left(\mu^{\mathtt{S}}\right)\left(1-q^{\mathtt{A}}\right) + \tau\left(\mu^{\mathtt{A}}\right) q^{\mathtt{A}}\right],$$

where the minimum and maximum are obtained by letting $\tilde{q} = 0$ and $\tilde{q} = 1$, respectively. Then, rearranging $\tau\left(\mu^{\mathtt{S}}\right)\left(1-q^{\mathtt{A}}\right)+\tau\left(\mu^{\mathtt{A}}\right) q^{\mathtt{A}} < 0$ yields the result, which guarantees $\tau < 0$. □

A.9 Proof of Proposition 4

Proof. First, by the assumptions that $\tau_k\left(\mu^{\mathtt{S}}\right) > \tau_k\left(\mu^{\mathtt{A}}\right)$ and $q_k^{\mathtt{S}} = 1 - q_k^{\mathtt{A}}$, observe that τ_k, the true ATE in study k, is decreasing in $q_k^{\mathtt{A}}$:

$$\begin{aligned}
\tau_k &= q_k^{\mathtt{S}} \tau_k\left(\mu^{\mathtt{S}}\right) + q_k^{\mathtt{A}} \tau_k\left(\mu^{\mathtt{A}}\right) \\
&= \tau_k\left(\mu^{\mathtt{S}}\right) - q_k^{\mathtt{A}}\left(\tau_k\left(\mu^{\mathtt{S}}\right) - \tau_k\left(\mu^{\mathtt{A}}\right)\right).
\end{aligned}$$

Next, by $\hat{\tau}_k = \tau_k + \varepsilon_k$, letting H denote ε_k's CDF leads to

$$\begin{aligned}
\Pr\left(\hat{\tau}_k > 0 \middle| q_k^{\mathtt{A}} = q\right) &= \Pr\left(\varepsilon_k > -\tau_k \middle| q_k^{\mathtt{A}} = q\right) \\
&= 1 - H\left(q\left(\tau_k\left(\mu^{\mathtt{S}}\right) - \tau_k\left(\mu^{\mathtt{A}}\right)\right) - \tau_k\left(\mu^{\mathtt{S}}\right)\right).
\end{aligned}$$

Because H is an increasing function, $\Pr\left(\hat{\tau}_k > 0 | q_k^{\mathtt{A}} = q\right)$ is decreasing in q. Then, by the assumption of $\Pr\left(\hat{\tau}_k > 0 | q_k^{\mathtt{A}} = q, k \in S^{\mathtt{S}}\right) = \Pr\left(\hat{\tau}_k > 0 | q_k^{\mathtt{A}} = q, k \in S^{\mathtt{A}}\right) = \Pr\left(\hat{\tau}_k > 0 | q_k^{\mathtt{A}} = q\right)$

and by the law of total probabilities, we obtain

$$\begin{aligned}
\Pr\left(\hat{\tau}_k > 0 \middle| k \in S^{\mathtt{S}}\right) &= \int_0^1 \Pr\left(\hat{\tau}_k > 0 \middle| q_k^{\mathtt{A}} = q, k \in S^{\mathtt{S}}\right) \mathrm{d}G^{\mathtt{S}}(q) \\
&= \int_0^1 \Pr\left(\hat{\tau}_k > 0 \middle| q_k^{\mathtt{A}} = q\right) \mathrm{d}G^{\mathtt{S}}(q), \text{ and, similarly,} \\
\Pr\left(\hat{\tau}_k > 0 \middle| k \in S^{\mathtt{A}}\right) &= \int_0^1 \Pr\left(\hat{\tau}_k > 0 \middle| q_k^{\mathtt{A}} = q\right) \mathrm{d}G^{\mathtt{A}}(q).
\end{aligned}$$

Because we have shown that $\Pr\left(\hat{\tau}_k > 0 | q_k^{\mathtt{A}} = q\right)$ is decreasing in q, $-\Pr\left(\hat{\tau}_k > 0 | q_k^{\mathtt{A}} = q\right)$ is increasing. Thus, Assumption 3' that $G^{\mathtt{A}}$ is FOSD over $G^{\mathtt{S}}$ implies

$$-\int_0^1 \Pr\left(\hat{\tau}_k > 0 \middle| q_k^{\mathtt{A}} = q\right) \mathrm{d}G^{\mathtt{A}}(q) > -\int_0^1 \Pr\left(\hat{\tau}_k > 0 \middle| q_k^{\mathtt{A}} = q\right) \mathrm{d}G^{\mathtt{S}}(q),$$

which is equivalent to $\Pr\left(\hat{\tau}_k > 0 | k \in S^{\mathtt{S}}\right) > \Pr\left(\hat{\tau}_k > 0 | k \in S^{\mathtt{A}}\right)$. □

B Existence and Uniqueness of the Cutpoint

Here, we discuss details on the existence of equilibria and the conditions under which uniqueness holds. Below, uniqueness means the uniqueness of the cutoff strategies $\overline{z}^{\mathtt{E}}, \underline{z}^{\mathtt{E}}$ in disaster periods, where $\mathtt{E} \in \{\mathtt{S}, \mathtt{A}_1, \mathtt{A}_2\}$. Proposition 3 above shows that there are regions of equilibrium multiplicity in the sense that two of the three strategy profiles $(\mathtt{S}, \mathtt{A}_1, \mathtt{A}_2)$ coexist even if the equilibrium cutpoints are unique.

B.1 Equilibrium Existence

Proposition 5. *An equilibrium exists in the game.*

Proof. Fix candidate equilibrium strategies $\mathtt{E} \in \{\mathtt{S}, \mathtt{A}_1, \mathtt{A}_2\}$. In any t such that $D_t = 0$, the expected payoffs from war and peace are fixed regardless of the case $(\mathtt{S})$, $(\mathtt{A}_1)$ or $(\mathtt{A}_2)$. It is easy to see that there must be an optimal action (either fighting or not fighting) whenever

$D_t = 0$.

Thus, consider an arbitrary subgame starting in period t such that $D_t = 1$. To define the subgame as an ordered normal form game and apply the results by Milgrom and Roberts (1990), consider one of the players' cutoff strategies in reverse order. That is, let Player 1's cutoff strategy be $\zeta_1 = 1 - \overline{z}^{\mathtt{E}}(\mu)$ for arbitrary case $(\mathtt{S})$, $(\mathtt{A}_1)$ or $(\mathtt{A}_2)$, so that a greater ζ_1 indicates a higher willingness to fight. Denote Player 2's by $\zeta_2 = \underline{z}^{\mathtt{E}}(\mu)$.

Denote the continuation payoffs in case $(\mathtt{E})$ as a function of these cutoffs, i.e., $\mathcal{V}_i^{\mathtt{E}}(\zeta_1, \zeta_2)$. By the assumption that F is continuous and by the expressions of $\mathcal{V}_i^{\mathtt{E}}(\zeta_1, \zeta_2)$ in Propositions 1 and 2, they are continuous in the cutoff strategies on $[0,1]^2$. By the extreme value theorem, for the other player j's cutoff, the best response $BR_i^{\mathtt{E}}(\zeta_j) \equiv \arg\max_{\zeta_i \in [0,1]} \mathcal{V}_i^{\mathtt{E}}(\zeta_1, \zeta_2)$ is nonempty and compact.

Because $[0,1]$ is a compact lattice, each best response correspondence has maximum and minimum elements. By Lemma 3, we know that, for the opponent's cutoffs $\zeta_j' \geq \zeta_j''$, we have

$$\max BR_i^{\mathtt{E}}\left(\zeta_j'\right) \geq \max BR_i^{\mathtt{E}}\left(\zeta_j''\right) \quad \text{and} \quad \min BR_i^{\mathtt{E}}\left(\zeta_j'\right) \geq \min BR_i^{\mathtt{E}}\left(\zeta_j''\right).$$

Then, there exist extremal equilibria that are found by iterating the players' best-response functions starting from (i) the case where players always fight when $D_t = 1$ and (ii) the case where they never fight when $D_t = 1$. □

B.2 Condition for Optimal Cutoff Uniqueness

The result below shows that there exists a unique equilibrium when the density of Z_t is not too large around the equilibrium cutoffs.

Proposition 6. *Suppose*

$$f\left(\overline{z}^{E}\right) f\left(\underline{z}^{E}\right) < \min \left\{ \begin{array}{l} \left(\dfrac{\left(1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{S},\underline{z}^{S}\right)\right)\right)\right)\left(\delta\overline{\mathcal{V}}-c\theta\right)}{\delta\pi\sqrt{\delta\mathcal{V}_1^{S}-\underline{z}^{S}\left(\delta\overline{\mathcal{V}}-c\theta\right)}\sqrt{\delta\mathcal{V}_2^{S}-\left(1-\overline{z}^{S}\right)\left(\delta\overline{\mathcal{V}}-c\theta\right)}} \right)^2, \\ \left\{ \left(\dfrac{\left(1-\delta\pi\Delta_F\left(\overline{z}^{A},\underline{z}^{A}\right)\right)\left(\delta\overline{\mathcal{V}}-c\theta\right)}{\delta\pi\sqrt{\delta\mathcal{V}_1^{A}-\underline{z}^{A}\left(\delta\overline{\mathcal{V}}-c\theta\right)}\sqrt{\delta\mathcal{V}_2^{A}-\left(1-\overline{z}^{A}\right)\left(\delta\overline{\mathcal{V}}-c\theta\right)}} \right)^2 \right\}_{A\in\{A_1,A_2\}} \end{array} \right\},$$

where $E \in \{S, A_1, A_2\}$*. Then, the equilibrium cutoffs are uniquely determined.*

Proof. We want to show that the mapping from an optimal cutpoint to an updated cutpoint, via the other player's best response to the former, is a contraction mapping so that the extremal equilibria converge to the unique fixed point. Consider Player 1 in the case (S). First, it is easy to see $\frac{\mathrm{d}\overline{z}^{\mathtt{S}}}{\mathrm{d}\mathcal{V}_1^{\mathtt{S}}} = \frac{\delta}{\theta\left[\delta\frac{2-\pi}{1-\delta}-c\right]}$. Next, take Player 2's cutoff $\underline{z}^{\mathtt{S}}$ as given. While $\mathcal{V}_1^{\mathtt{S}}$ is a function of both $\overline{z}^{\mathtt{S}}$ and $\underline{z}^{\mathtt{S}}$ and these thresholds depend on each other, since $\overline{z}^{\mathtt{S}}$ is the optimal cutpoint given $\underline{z}^{\mathtt{S}}$, the envelop theorem implies $\frac{\mathrm{d}\mathcal{V}_1^{\mathtt{S}}}{\mathrm{d}\underline{z}^{\mathtt{S}}} = \frac{\partial\mathcal{V}_1^{\mathtt{S}}}{\partial\underline{z}^{\mathtt{S}}}$, which is given by

$$\frac{\partial\mathcal{V}_1^{\mathtt{S}}}{\partial\underline{z}^{\mathtt{S}}} = -\pi f\left(\underline{z}^{\mathtt{S}}\right)\cdot\frac{\delta\mathcal{V}_1^{\mathtt{S}}-\underline{z}^{\mathtt{S}}\left(\delta\overline{\mathcal{V}}-c\theta\right)}{1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\right)\right)}.$$

Then, denoting $\overline{z}^{\mathtt{S}}$ as a function of $\underline{z}^{\mathtt{S}}$, we obtain

$$\begin{aligned} \frac{\mathrm{d}\overline{z}^{\mathtt{S}}\left(\underline{z}^{\mathtt{S}}\right)}{\mathrm{d}\underline{z}^{\mathtt{S}}} &= \frac{\mathrm{d}\overline{z}^{\mathtt{S}}}{\mathrm{d}\mathcal{V}_1^{\mathtt{S}}}\cdot\underbrace{\frac{\mathrm{d}\mathcal{V}_1^{\mathtt{S}}}{\mathrm{d}\underline{z}^{\mathtt{S}}}}_{=\frac{\partial\mathcal{V}_1^{\mathtt{S}}}{\partial\underline{z}^{\mathtt{S}}}} \\ &= -f\left(\underline{z}^{\mathtt{S}}\right)\cdot\frac{\delta\pi\left(\delta\mathcal{V}_1^{\mathtt{S}}-\underline{z}^{\mathtt{S}}\left(\delta\overline{\mathcal{V}}-c\theta\right)\right)}{\left(1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\right)\right)\right)\left(\delta\overline{\mathcal{V}}-c\theta\right)}. \end{aligned}$$

Analogously, for Player 2, we have

$$\begin{aligned}\frac{\mathrm{d}\underline{z}^{\mathtt{S}}\left(\overline{z}^{\mathtt{S}}\right)}{\mathrm{d}\overline{z}^{\mathtt{S}}} &= \frac{\mathrm{d}\underline{z}^{\mathtt{S}}}{\mathrm{d}\mathcal{V}_2^{\mathtt{S}}}\cdot\frac{\partial\mathcal{V}_2^{\mathtt{S}}}{\partial\overline{z}^{\mathtt{S}}} \\ &= -f\left(\overline{z}^{\mathtt{S}}\right)\cdot\frac{\delta\pi\left(\delta\mathcal{V}_2^{\mathtt{S}}-\left(1-\overline{z}^{\mathtt{S}}\right)\left(\delta\overline{\mathcal{V}}-c\theta\right)\right)}{\left(1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\right)\right)\right)\left(\delta\overline{\mathcal{V}}-c\theta\right)}.\end{aligned}$$

By the contraction mapping theorem, the optimal cutoff given the other player's cutoff converges to a unique fixed point when

$$\begin{aligned}\frac{\mathrm{d}\overline{z}^{\mathtt{S}}\left(\underline{z}^{\mathtt{S}}\right)}{\mathrm{d}\underline{z}^{\mathtt{S}}}\cdot\frac{\mathrm{d}\underline{z}^{\mathtt{S}}\left(\overline{z}^{\mathtt{S}}\right)}{\mathrm{d}\overline{z}^{\mathtt{S}}} &< 1 \\ f\left(\overline{z}^{\mathtt{S}}\right)f\left(\underline{z}^{\mathtt{S}}\right) &< \left(\frac{\left(1-\delta\left(1-\pi\left(1-\Delta_F\left(\overline{z}^{\mathtt{S}},\underline{z}^{\mathtt{S}}\right)\right)\right)\right)\left(\delta\overline{\mathcal{V}}-c\theta\right)}{\delta\pi\sqrt{\delta\mathcal{V}_1^{\mathtt{S}}-\underline{z}^{\mathtt{S}}\left(\delta\overline{\mathcal{V}}-c\theta\right)}\sqrt{\delta\mathcal{V}_2^{\mathtt{S}}-\left(1-\overline{z}^{\mathtt{S}}\right)\left(\delta\overline{\mathcal{V}}-c\theta\right)}}\right)^2.\end{aligned}$$

Finally, recall that Assumption 1 imposes a unique optimal action in no-disaster periods. Consequently, when the above inequality holds, there exists a unique equilibrium in the game. We can derive the conditions for cases ($\mathtt{A}_1$) and ($\mathtt{A}_2$) in the same manner. □

C Allowing Bargaining Before Deciding Whether to Fight

For simplicity, the model in the main text assumes that players do not bargain over resources to avoid an inefficient conflict. Here, I show that the key insights of the model hold even when we allow players to transfer resources. Now assume that, in each period, players have an arbitrary bargaining protocol between Nature's move and their decision-making about war. During this bargaining process, they can transfer the resources they possess in the given period. Namely, in a no-disaster period, each player can offer up to θ to the other. In a disaster period, Players 1 and 2 can transfer up to $z_t\theta$ and $(1-z_t)\theta$, respectively, where z_t is the realized value of Z_t at period t. Hence, a player can obtain a flow payoff up to θ

(disaster period) or 2θ (no-disaster period) as a result of bargaining.

An important observation here is that players cannot commit to future transfers of resources. Even if a player attempts to avoid conflict by announcing a menu of long-term transfers in an arbitrary bargaining protocol, such a promise is not credible because the proposer cannot resist the temptation to renege on it when she has a great military advantage with an extreme value of z_t. Namely, forgoing a highly favorable state variable (i.e., large and small z_t for Players 1 and 2, respectively) because of a past promise would violate subgame perfection. This is in line with the core idea of other commitment-problem models in international relations and political economy (Fearon, 1995; Powell, 2006, 2004).

In summary, the timeline for each period of the game with bargaining is as follows.

1. Nature draws D_t and Z_t.
2. Players observe D_t and Z_t, and bargain over the division of the pie (θ when $D_t = 1$ and 2θ when $D_t = 0$) via an arbitrary protocol.
3. Players simultaneously decide whether or not to attack the other.
    - If either player fights, the game ends
    - Otherwise, the game proceeds to the next period, $t + 1$, where the flow payoff for t is determined by the bargaining phase

Note that we assume commitment power *within a period*. Namely, after the decision not to fight in period t, players cannot renegotiate the flow payoff in t.

C.1 Opportunistic Motive

It is intuitive that the possibility of bargaining should expand the prospect of peace because (i) the militarily disadvantaged side can offer the remaining resources to the advantaged side and (ii) the side that is more robust to disasters can compensate the vulnerable player in a

no-disaster period in order to avoid an inefficient conflict. Thus, rather than constructing a specific equilibrium with bargaining, we are interested in whether there can be a completely peaceful equilibrium in which war never occurs on its path.

Definition 1. A *peaceful equilibrium* is a subgame perfect equilibrium in which no player attacks the other side on the equilibrium path.

If war is still possible on the equilibrium path, then we can say that the key observations in the main text still hold. The following result shows that the opportunistic motive for conflict remains, as long as the cost of war is sufficiently small and Z_t can take extreme values close enough to zero or one.

Proposition 7. *Suppose (i) the cost of conflict, c, is sufficiently small and (ii) Z_t has full support over $[0,1]$. Then, a peaceful equilibrium does not exist in the game with bargaining, where war occurs due to an opportunistic aggression at t such that $D_t = 1$.*

Proof. Suppose to the contrary that there exists a peaceful equilibrium. To derive a contradiction, focus on Player 1's decision-making when $D_t = 1$. Denote her continuation value in the peaceful equilibrium by $\mathcal{V}_1^{\texttt{peace}}$. First, observe $\mathcal{V}_1^{\texttt{peace}} < \overline{\mathcal{V}}$. To see this, recall that $\overline{\mathcal{V}} = \frac{\theta(2-\pi)}{1-\delta}$ is the long-term payoff that the victor in war receives, where she is assumed to dominate all the resources (i.e., $Z_t\theta + (1-Z_t)\theta = \theta$ or $\theta + \theta = 2\theta$) in the society for an infinite number of periods. This quantity also represents the pie to be divided via bargaining between players. Thus, $\mathcal{V}_1^{\texttt{peace}} = \overline{\mathcal{V}}$ would imply Player 2 receives a flow payoff of zero every period. However, observe that the minmax payoff that Player 2 can secure in each disaster and no-disaster period is, respectively,

$$\mathcal{W}_2^{\texttt{d}}(z_t) = (1-z_t)\left(\theta(1-c) + \delta\overline{\mathcal{V}}\right) > 0 \quad \text{or} \quad \mathcal{W}_2^{\neg\texttt{d}} = \frac{1}{2}\left(2\theta(1-c) + \delta\overline{\mathcal{V}}\right) > 0.$$

The strictly positive minmax payoffs of Player 2 imply that, to sustain a peaceful equilibrium, Player 1 has to offer Player 2 a strictly positive amount of resources in each period. Otherwise,

Player 2 would unilaterally deviate to fighting, and peace would break down. This observation establishes $\mathcal{V}_1^{\texttt{peace}} < \overline{\mathcal{V}}$.

We consider Player 1's decision-making in a disaster period to check if a peaceful equilibrium exists based on the one-shot deviation principle. By maintaining peace under an arbitrary bargaining protocol, Player 1 can earn at most $z_t\theta + (1-z_t)\theta + \delta\mathcal{V}_1^{\texttt{peace}}$ in a disaster period. Then, rearranging $\theta + \delta\mathcal{V}_1^{\texttt{peace}} \geq \mathcal{W}_1^{\texttt{d}}(z_t) = z_t\left(\theta(1-c) + \delta\overline{\mathcal{V}}\right)$ yields

$$\theta(1-z_t) + \delta\mathcal{V}_1^{\texttt{peace}} \geq z_t\left(\delta\overline{\mathcal{V}} - \theta c\right).$$

Observe that, as z_t approaches one, the left-hand side and the right-hand side converge to $\delta\mathcal{V}_1^{\texttt{peace}}$ and $\delta\overline{\mathcal{V}} - \theta c$, respectively. Then, by $\mathcal{V}_1^{\texttt{peace}} < \overline{\mathcal{V}}$ and by the assumption of full support, if c is sufficiently small, there exists a value of z_t above which the inequality does not hold and Player 1 attacks Player 2 in a disaster period. This violates the one-shot deviation principle, which is a contradiction with the assumption that a peaceful equilibrium exists. Similarly, there exist values of z_t small enough for Player 2 to fight even with the opportunity to bargain over the resources. □

C.2 Preemptive Motive

Next, we are interested in whether the insights in the second, asymmetric case still hold under bargaining. The following result shows that, when one of the players is disproportionately vulnerable to future disasters, the preemptive motive for conflict remains in the game with bargaining.

Proposition 8. *If the discount factor, δ, is sufficiently large, there exist values of μ under which a peaceful equilibrium does not exist in the game with bargaining, where war occurs due to a preemptive aggression at t such that $D_t = 0$.*

Proof. Focus on Player 1's decision-making in any subgame that starts from a period such

that $D_t = 0$. We apply Powell's inefficiency condition (Powell, 2004). Proposition 1 of Powell (2004) shows that there cannot be an efficient equilibrium in a stochastic game with complete information and bargaining when a player's minmax payoff is greater than the difference between (i) the pie to be divided and (ii) the other player's largest concession in the current state and her (discounted) expected minmax payoff. To represent the inefficiency condition in the context of the current model, let $\mathcal{W}_2(d, z)$ be Player 2's war payoff when $D_t = d$ and $Z_t = z$.

Then, we can express the inefficiency condition when $D_t = 0$ as

$$\underbrace{\mathcal{W}_1^{\neg \mathtt{d}}}_{\text{P1's minmax payoff } (D_t=0)} > \underbrace{2\theta + \delta\overline{\mathcal{V}}}_{\text{Pie to be divided}} - \left(\underbrace{0}_{\text{P2's largest concession}} + \underbrace{\delta \mathbb{E}_{D_t, Z_t}\left[\mathcal{W}_2(D_t, Z_t)\right]}_{\text{P2's (discounted) expected minmax}} \right)$$

This leads to

$$\begin{aligned}
\mathcal{W}_1^{\neg \mathtt{d}} &> 2\theta + \delta\overline{\mathcal{V}} - \delta\left((1-\pi)\mathcal{W}_2^{\neg \mathtt{d}} + \pi \mathbb{E}_{Z_t}\left[\mathcal{W}_2^{\mathtt{d}}(Z_t)\right]\right) \\
&= 2\theta + \delta\overline{\mathcal{V}} - \delta\left((1-\pi)\mathcal{W}_2^{\neg \mathtt{d}} + \pi \int_0^1 (1-z)\left(\theta(1-c) + \delta\overline{\mathcal{V}}\right) f(z)\mathrm{d}z\right) \\
&= 2\theta + \delta\overline{\mathcal{V}} - \delta\left((1-\pi)\mathcal{W}_2^{\neg \mathtt{d}} + \pi(1-\mu)\left(\theta(1-c) + \delta\overline{\mathcal{V}}\right)\right).
\end{aligned}$$

Given that $\mathcal{W}_1^{\neg \mathtt{d}} = \mathcal{W}_2^{\neg \mathtt{d}} = \frac{1}{2}\left(2\theta(1-c) + \delta\overline{\mathcal{V}}\right)$, rearranging the inequality yields

$$\mu < \frac{\frac{1}{2}\delta\overline{\mathcal{V}}\left(\delta(1+\pi) - 1\right) - \theta\left(2 - (1+\delta)(1-c)\right)}{\delta\pi\left(\theta(1-c) + \delta\overline{\mathcal{V}}\right)}.$$

Observe that the denominator is positive. The numerator is positive when

$$(1+\delta)(1-c) + \frac{\delta}{1-\delta}\frac{2-\pi}{2}\left(\delta(1+\pi) - 1\right) > 2.$$

Thus, for any values of $\pi, c \in (0,1)$, there exist values of μ under which peace breaks down due to the preemptive motive when δ is sufficiently large. □

D Categorizing Studies in Burke et al. (2024)

Table 2 below classifies the studies collected by Burke et al. (2024) based on the asymmetric-vulnerability theory. The combination of group types of a study is classified as "Asymmetric" if it relies on datasets of conflicts between government forces and non-state actors. It is said to be "Symmetric" if the study uses datasets of conflicts either (i) between different non-state actors or (ii) between different government factions. If the datasets used in a study contain conflict events between both "Asymmetric" and "Symmetric" groups, it is classified as "Various." The theory predicts that studies with "Symmetric" and "Asymmetric" groups should have a positive and negative estimated effect of climate events on armed conflict, respectively. Because the theory is agnostic about the "Various" cases, the prediction of such studies is said to be "Indeterminate." The column "Effect in Burke et al." indicates the sign of the estimated effect of each study reported in Burke et al. (2024).

Table 2: Meta-analysis categorization

Paper	Conflict type	Climate event	Conflict data	Climate data	(A)symmetry	Effect in Burke et al.	Predicted sign
Almer et al. (2017)	Riots	Drought	Social Conflict Analysis Database (SCAD)	Standardized Precipitation-Evapotranspiration Index (SPEI)	Symmetric	Positive	Positive
Baysan et al. (2019)	Gang violence	Temperature	Homicide data from Mexico's Bureau of Statistics (INEGI)	Global (land) precipitation and temperature, University of Delaware	Symmetric	Positive	Positive
Bergholt and Lujala (2012)	Civil war	Storm	UCDP/PRIO	EM-DAT	Asymmetric	Negative	Negative
Bohlken and Sergenti (2010)	Riots	Rainfall deviation	Hindu-Muslim riots data from Varshney and Wilkinson[22]	Global Precipitation Climatology Project (GPCP)	Symmetric	Positive	Positive
Böhmelt et al. (2014)	Water conflict	Temperature deviation	Water-related intrastate conflict and cooperation database by Bernauer et al. (2012)	University of Delaware data	Various	Positive	Indeterminate
Bollfrass and Shaver (2015)	Civil war	Temperature	PRIO Conflict Site	PRIO-GRID	Asymmetric	Negative	Negative
Bounadi (2018)	Casualty	Temperature	UCDP Georeferenced Event Dataset (GED)	Asia Land Information System (LIS) Framework of the Hydrological Sciences Laboratory at the NASA GSFC	Various	Positive	Indeterminate

[22] https://doi.org/10.3886/ICPSR04342.v1

Table 2 (continued)

Paper	Conflict type	Climate event	Conflict data	Climate data	(A)symmetry	Effect in Burke et al.	Predicted sign
Breckner and Sunde (2019)	Conflict events	Temperature deviation	Armed Conflict Location and Event Data (ACLED)	European Center for Medium Run Weather Forecast (ECMWF)	Various	Positive	Indeterminate
Brückner and Ciccone (2011)	Institutional change (democratization)	Rainfall	POLITY IV	GPCP	Various	Negative	NA
Buhaug (2010)	Civil war	Temperature	UCDP/PRIO	Climate Research Unit (CRU) time-series data, University of East Anglia	Asymmetric	Negative	Negative
Burke and Leigh (2010)	Institutional change (democratization)	Temperature deviation	POLITY IV	Tyndall Center for Climate Change Research (TYN) country-average data	Various	Positive	NA
Burke (2012)	Institutional change (leader)	Temperature	Archigos	Dell et al. (2012)	Various	Positive	NA
Burke et al. (2009)	Civil war	Temperature	UCDP/PRIO	CRU	Asymmetric	Positive	Negative
Caruso et al. (2017)	Casualty	Rainfall	UN OCHA in Juba	World Food Program and the South Sudan National Bureau of Statistics	Various	Positive	Indeterminate

Table 2 (continued)

Paper	Conflict type	Climate event	Conflict data	Climate data	(A)symmetry	Effect in Burke et al.	Predicted sign
Caruso et al. (2016)	Conflict events	Temperature deviation	United Nations Support Facility for Indonesian Recovery (UNSFIR) data	Global Historical Climatology Network (GHCN) data of the National Oceanic and Atmospheric Administration (NOAA)	Various	Positive	Indeterminate
Couttenier and Soubeyran (2014)	Civil war	Drought	UCDP/PRIO	Palmer Drought Severity Index (PDSI)	Asymmetric	Positive	Negative
Craig et al. (2021)	Terrorism	Temperature	Global Terrorism Database (GTD)	GHCN data	Asymmetric	Positive	Negative
Dell et al. (2012) I	Institutional change (leader)	Temperature deviation	Archigos	University of Delaware (Terrestrial Air Temperature) data	Various	Positive	NA
Dell et al. (2012) II	Civil war	Temperature deviation	UCDP/PRIO	University of Delaware (Terrestrial Air Temperature) data	Asymmetric	Negative	Negative
Fetzer (2020)	Conflict events	Rainfall	South Asia Terrorism Portal (SATP)	Tropical Rainfall Measuring Mission (TRMM) satellite data	Various	Positive	Indeterminate
Fjelde and von Uexkull (2012)	Non-state conflict	Rainfall deviation	UCDP GED	GPCP	Symmetric	Positive	Positive

Table 2 (continued)

Paper	Conflict type	Climate event	Conflict data	Climate data	(A)symmetry	Effect in Burke et al.	Predicted sign
Gatti et al. (2021)	Social conflict	Growing season rainfall	Indonesia's National Violence Monitoring System (NVMS) data	Climate Hazards Group InfraRed Precipitation with Station (CHIRPS) data, UC Santa Barbara	Symmetric	Positive	Positive
Gerling (2017) I	Protest duration	Drought	Social Conflict in Analysis Database (SCAD)	Miguel et al. (2004), Miguel and Satyanath (2011), and Ciccone (2011), based on GPCP	Various	Positive	NA
Gerling (2017) II	Coup	Drought	Powell and Thyne (2011)	Miguel et al. (2004), Miguel and Satyanath (2011), and Ciccone (2011), based on GPCP	Symmetric	Negative	Positive
Guardado and Pennings (2025)	Conflict events	Temperature	Iraq Body Count (IBC) dataset; GTD; World Incident Tracking System (WITS); SIGACTS data	University of Delaware data	Various	Negative	Indeterminate
Harari and Ferrara (2018)	Conflict events	Drought	ACLED	SPEI	Various	Positive	Indeterminate
Helman et al. (2020)	Non-state conflict	Temperature deviation	UCDP GED	Climate Hazards Center InfraRed Temperature with Stations (CHIRTS) data	Symmetric	Negative	Positive

Table 2 (continued)

Paper	Conflict type	Climate event	Conflict data	Climate data	(A)symmetry	Effect in Burke et al.	Predicted sign
Hendrix and Salehyan (2012)	Social conflict	Rainfall deviation	SCAD	GPCP	Symmetric	Positive	Positive
Hidalgo et al. (2010)	Non-state conflict	Rainfall deviation	Land invasion data from Brazil's Pastoral Land Commission (Comissão Pastoral da Terra, CPT) and Dataluta (Banco de Dados da Luta pela Terra)	Rainfall data from the Brazilian National Water Agency	Symmetric	Positive	Positive
Hodler and Raschky (2014)	Conflict events	Rainfall deviation	UCDP GED	CRU data	Various	Positive	Indeterminate
Kim (2016)	Coup	Temperature deviation	Powell and Thyne (2011)	Dell et al. (2012)	Symmetric	Positive	Positive
König et al. (2017)	Conflict events	Rainfall	ACLED	Global Precipitation and Climatology Centre (GPCC) data	Various	Positive	Indeterminate
Landis (2014) I	Civil war	Temperature deviation	UCDP/PRIO	NCEP-NCAR Reanalysis Monthly Means data of the NOAA	Asymmetric	Negative	Negative
Landis (2014) II	Non-state conflict	Temperature deviation	UCDP Non-state Conflict data, SCAD	NCEP-NCAR Reanalysis Monthly Means data	Symmetric	Positive	Positive
Landis et al. (2017)	Conflict events	Temperature	ACLED	CRU	Various	Positive	Indeterminate

Table 2 (continued)

Paper	Conflict type	Climate event	Conflict data	Climate data	(A)symmetry	Effect in Burke et al.	Predicted sign
Levy et al. (2005)	Civil war	Drought	UCDP/PRIO	Weighted Anomaly Standardized Precipitation (WASP) index	Asymmetric	Positive[23]	Negative
Linke and Ruether (2021)	Civil war	Temperature	Integrated Crisis Early Warning System (ICEWS) data	ERA5 surface air temperature data from the Copernicus Program, European Center for Medium-range Weather Forecasts (ECMWF)	Asymmetric	Positive	Negative
Maystadt et al. (2015)	Conflict events	Temperature deviation	ACLED	CRU	Various	Positive	Indeterminate
Miguel et al. (2004)	Civil war	Rainfall deviation	UCDP/PRIO	GPCP	Asymmetric	Positive	Negative
O'Loughlin et al. (2012)	Conflict events	Temperature deviation	ACLED	CRU	Various	Positive	Indeterminate
Salehyan and Hendrix (2014)	Civil war	Drought	UCDP/PRIO	PDSI	Asymmetric	Negative[24]	Negative
Shaver and Bollfrass (2023)	IED attacks	Temperature	IED attacks in SIGACT data	U.S. National Climatic Data Center	Various	Positive	Indeterminate

[23]Although Table 1 of Burke et al. (2024) reports a negative estimated effect of precipitation on conflict, the interpretation here is that a climate disaster (i.e., presence of drought) has a positive effect on the risk of civil war. Thus, this study does not fit my theory. Figure 5 of Burke et al. (2024) shows a positive effect.

[24]Salehyan and Hendrix (2014) predict that droughts lead to a decline in highly organized armed conflict. A smaller PDSI score indicates a drought. Thus, while Table 1 in Burke et al. (2024) presents a positive estimated effect, the findings of Salehyan and Hendrix (2014) imply a negative relationship between a climate disaster (the presence of drought) and conflict.

Table 2 (continued)

Paper	Conflict type	Climate event	Conflict data	Climate data	(A)symmetry	Effect in Burke et al.	Predicted sign
Smith (2014)	Non-state conflict	Rainfall deviation	SCAD	GPCC	Symmetric	Negative	Positive
Theisen et al. (2011)	Civil war	Drought	UCDP/PRIO	GPCC	Asymmetric	Negative	Negative
von Uexkull et al. (2016)	Conflict events	Drought	UCDP GED	SPEI	Various	Positive	Indeterminate
Wischnath and Buhaug (2014)	Civil war	Temperature	PRIO Conflict Site	University of Delaware data	Asymmetric	Negative	Negative
Yeeles (2015)	Non-state conflict	Temperature	PRIO's Urban Social Disturbance in Africa and the Middle East (USDA) dataset	University of Delaware data	Various	Positive	Indeterminate

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